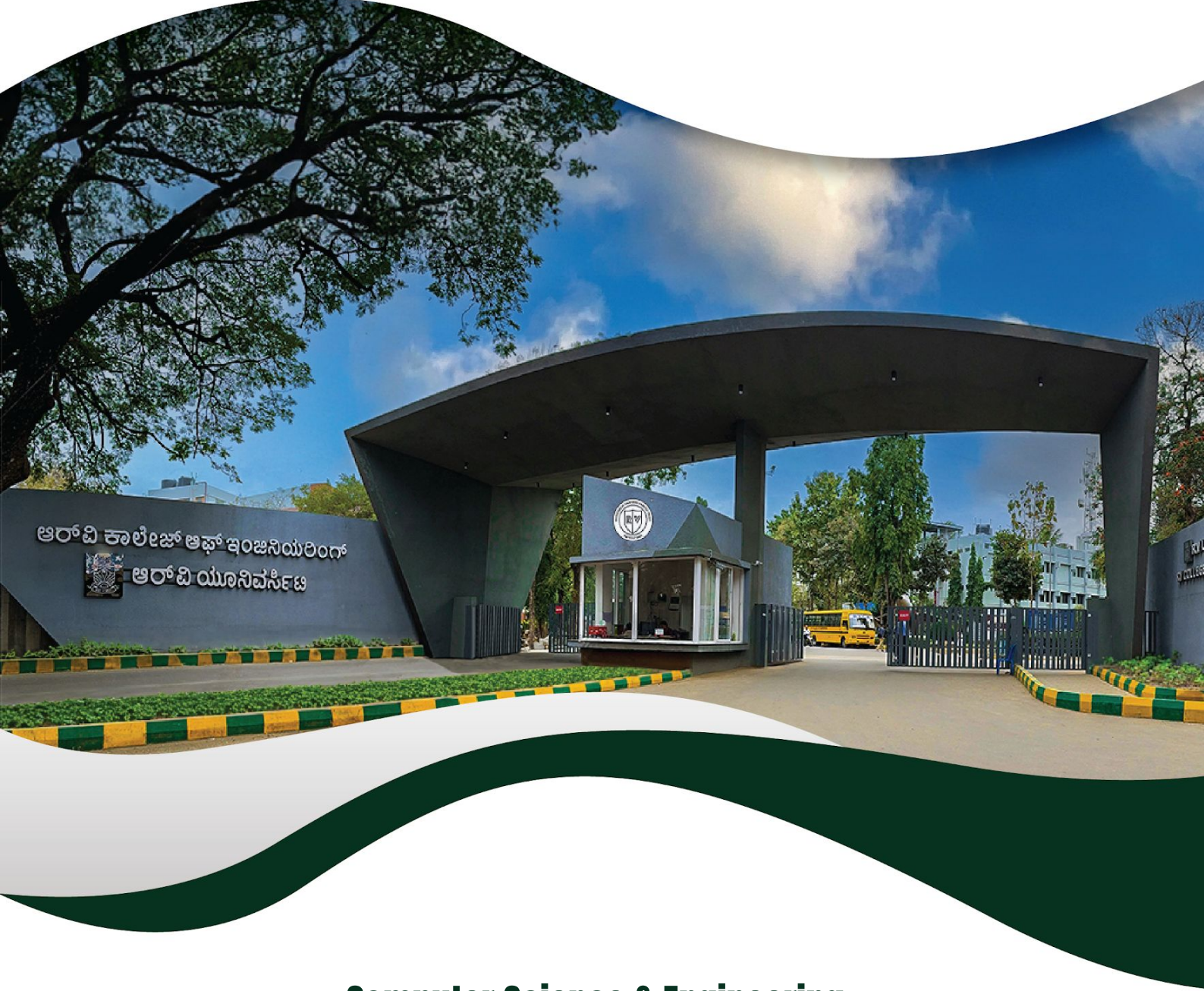




**RV College of
Engineering®**



Computer Science & Engineering

Bachelor of Engineering (B.E)

Scheme And Syllabus Of VII & VIII Semester
(2022 Scheme)

B.E. Programs : AI, AS, BT, CD, CH, CS, CV, CY, EC, EE, EI, ET, IM, IS, ME.

M. Tech (13) MCA, M.Sc. (Engg.)

Ph.D. Programs : All Departments are recognized as Research Centres by VTU Except AI & AS

2025



RV College of
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SINCE
1963

NAAC
GRADE **A+**
ACCREDITED INSTITUTION

SHAPING VISIONARIES, CRAFTING INNOVATORS



An autonomous institution affiliated with VTU, Belagavi, and approved by AICTE and UGC, New Delhi

- **RV SATELLITE-1 (RVSAT - 1)** India's First Microbiological Payload, launched by RVCE students in December 2024
- **NCC** - Best Institution Award of Karnataka and Goa Directorate (2019, 2022, 2023 & 2025)

₹50 Cr.

MoU with **TATA Technologies** to establish Centre for Invention, Innovation, Incubation & Training

Placements

₹92 LPA
Highest Package

250+
Recruitment Partners

92%
Placement Rate

₹10 LPA
Median Salary

Rankings

99th
NIRF Ranking 2024
in Engineering
Category

1501+
Times Higher
Education World
University
Rankings-2025

601+
Times Higher
Education Asia
University
Rankings-2025

NBA
UG and PG
programs
accredited
multiple times

IIRF 2025
Engineering ranking India
National - 01
State - 01
Zone - 01

NPTEL
Local Chapter Ranking 2025
National-09
State - 01

Research and Centres

37 Skill Labs

General Skills, Domain Skills, Interdisciplinary Skills

- Bosch Rexroth
- Mercedes Benz
- Morris Garage
- Boston AI System
- Toyota Kirloskar
- Vision Astra EV Academy

Outcome Based Education (OBE) emphasizing on experiential learning, research-based learning and skill-based learning

20

Centres of
Excellence

8

Centres of
Competence

80

Patents
filed

65

Patents
published

46

Patents
granted

32

Innovative & Cultural
activity teams

₹30 Cr.

Research and
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(last 5 years)

16 Research Centres
in niche areas

- Autonomous Vehicles
- Nanomaterials and Devices
- Hydrogen and Green Technology
- Extended Reality
- Healthcare
- E-mobility
- IoT and Sensors
- Smart Antenna System
- Computational Genomics
- Quantum Computing

Programs and Faculty

- 13 UG Programs;
1420 (2025) annual sanctioned intake
- 13 PG Programs
390 (2025) annual sanctioned intake
- Ph.D. Programs in all departments
- 500+ Faculty Members and Staff

CET / KEA Code: **E005**
COMED-K Code: **E095**

PGCET Code (MTech): **T857**
PGCET Code (MCA): **C463**

State-of-the-Art infrastructure

- Innovation Center
- Gymnasium
- Hostels
- Labs & Workshops

Alumni

36,000 +

Alumni across the globe

100+

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NUMBER OF ATTENDEES

650

San Jose

120

Seattle

Mr. Suresh Katta, RVCE alumnus & Chairman Emeritus of Saama Group, USA and Mr. Anil Kumble, RVCE alumnus & former Indian cricketer, were the Guests of Honour at the alumni meets held in San Jose & Seattle, USA.

Collaborations with International universities

125

MoUs with Industry, Institutions & Research Establishments



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Computer Science & Engineering

Bachelor of Engineering (B.E)

Scheme And Syllabus Of VII & VIII Semester
(2022 Scheme)

B.E. Programs : AI, AS, BT, CD, CH, CS, CV, CY, EC, EE, EI, ET, IM, IS, ME.

M. Tech (13) MCA, M.Sc. (Engg.)

Ph.D. Programs : All Departments are recognized as Research Centres by VTU Except AI & AS

2025



VISION

Leadership in Quality Technical Education,
Interdisciplinary Research & Innovation, with a Focus
on Sustainable and Inclusive Technology

MISSION

1. To deliver outcome based Quality education, emphasizing on experiential learning with the state of the art infrastructure.
2. To create a conducive environment for interdisciplinary research and innovation.
3. To develop professionals through holistic education focusing on individual growth, discipline, integrity, ethics and social sensitivity.
4. To nurture industry-institution collaboration leading to competency enhancement and entrepreneurship.
5. To focus on technologies that are sustainable and inclusive, benefiting all sections of the society.

QUALITY POLICY

Achieving Excellence in Technical Education, Research and Consulting through an Outcome Based Curriculum focusing on Continuous Improvement and Innovation by Benchmarking against the global Best Practices.

CORE VALUES

Professionalism, Commitment, Integrity, Team Work,
Innovation



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DEPARTMENT VISION

To achieve leadership in the field of Computer Science & Engineering by strengthening fundamentals and facilitating interdisciplinary sustainable research to meet the ever growing needs of the society.

DEPARTMENT MISSION

- To evolve continually as a centre of excellence in quality education in computers and allied fields.
- To develop state-of-the-art infrastructure and create environment capable for interdisciplinary research and skill enhancement.
- To collaborate with industries and institutions at national and international levels to enhance research in emerging areas.
- To develop professionals having social concern to become leaders in top-notch industries and/or become entrepreneurs with good ethics.

PROGRAM EDUCATIONAL OBJECTIVES(PEOs)

- PEO1:** Develop Graduates capable of applying the principles of mathematics, science, core engineering and Computer Science to solve real-world problems in interdisciplinary domains.
- PEO2:** To develop the ability among graduates to analyze and understand current pedagogical techniques, industry accepted computing practices and state-of-art technology.
- PEO3:** To develop graduates who will exhibit cultural awareness, teamwork with professional ethics, effective communication skills and appropriately apply knowledge of societal impacts of computing technology.
- PEO4:** To prepare graduates with a capability to successfully get employed in the right role /become entrepreneurs to achieve higher career goals or takeup higher education in pursuit of lifelong learning.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO	Description
PSO1	System Analysis and Design The student will be able to: 1. Recognize and appreciate the need of change in computer architecture, data organization and analytical methods in the evolving technology. 2. Learn the applicability of various systems software elements for solving design problems. 3. Identify the various analysis & design methodologies for facilitating development of high quality system software products with focus on performance optimization. 4. Display team participation, good communication, project management and document skills.
PSO2	Product Development The student will be able to: 1. Demonstrate the use of knowledge and ability to write programs and integrate them with the hardware/software products in the domains of embedded systems, databases/data analytics, network/web systems and mobile products. 2. Participate in planning and implement solutions to cater to business – specific requirements displaying team dynamics and professional ethics. 3. Employ state-of-art methodologies for product development and testing / validation with focus on optimization and quality related aspects.

**Lead Society: Institute of Electrical and Electronics Engineers
(IEEE)**

ABBREVIATIONS

Sl. No.	Abbreviation	Meaning
1.	VTU	Visvesvaraya Technological University
2.	BS	Basic Sciences
3.	CIE	Continuous Internal Evaluation
4.	SEE	Semester End Examination
5.	PE	Professional Core Elective
6.	GE	Global Elective
7.	HSS	Humanities and Social Sciences
8.	PY	Physics
9.	CM	Chemistry
10.	MA	Mathematics
11.	AS	Aerospace Engineering
12.	AI & ML	Artificial Intelligence & Machine Learning
13.	BT	Biotechnology
14.	CH	Chemical Engineering
15.	CS	Computer Science & Engineering
16.	CV	Civil Engineering
17.	EC	Electronics & Communication Engineering
18.	EE	Electrical & Electronics Engineering
19.	EI	Electronics & Instrumentation Engineering
20.	ET	Electronics & Telecommunication Engineering
21.	IM	Industrial Engineering & Management
22.	IS	Information Science & Engineering
23.	ME	Mechanical Engineering
24.	CD	Computer Science & Engineering(Data Science)
25.	CY	Computer Science & Engineering(Cyber Security)
26.	CI	Computer Science & Engineering(Artificial Intelligence & Machine Learning)

INDEX

VII Semester			
Sl. No.	Course Code	Course Title	Page No.
1.	HS271TA	Indian Knowledge System	1
2.	CS372IA	Parallel Architecture and GPU Programming	3
3.	CS373IA	Computer Graphics and Virtual Reality	6
4.	CS374TFX	Professional Core Elective-IV (Group-F)	10 - 23
5.	375TGX	Institutional Electives – II (Group G)	24 – 64
6.	CS376SI	Summer Internship	65
VIII Semester			
1.	CS481P	Major Project	67



Bachelor of Engineering in COMPUTER SCIENCE AND ENGINEERING VII SEMESTER													
Sl. No.	Course Code	Course Title	Credit Allocation				BoS	Category	Max Marks CIE		SEE Duration (H)	Max Marks SEE	
			L	T	P	Total			Theory	Lab		Theory	Lab
1	HS271TA	Indian Knowledge System	3	0	0	3	HS	Theory	100	****	3	100	****
2	CS372IA	Parallel Architecture and GPU Programming	3	0	1	4	CS	Theory + Practice	100	50	3	100	50
3	CS373IA	Computer Graphics and Virtual Reality	3	0	1	4	CS	Theory + Practice	100	50	3	100	50
4	CS374TFX	Professional Core Elective-IV Group-F	3	0	0	3	CS	Theory	100	****	3	100	****
5	XX375TGX	Institutional Electives – II (Group G)	3	0	0	3	CS	Theory	100	****	3	100	****
6	CS376SI	Summer Internship	0	0	3	3	CS	Internship	****	100	3	****	100
		Total	15	0	5	20							

Note: * Internship (8 weeks) is to be carried during the vacation after 6th semester and evaluation shall be conducted during 7th semester for 3 credits

VIII SEMESTER													
Sl. No.	Course Code	Course Title	Credit Allocation				BoS	Category	Max Marks CIE		SEE Duration (H)	Max Marks SEE	
			L	T	P	Total			Theory	Lab		Theory	Lab
1	CS481P	Major Project	0	0	12	12	CS	Project	****	100	3	****	100
		Total				12							



VII Semester PROFESSIONAL CORE ELECTIVE-IV GROUP-F		
Sl. No.	Course Code	Course Title
1	CS374TFA	Deep Learning
2	CS374TFB	Cyber Security for Industry 4.0 (Common to CS & IS)
3	CS374TFC	Application Delivery Controller and Virtualization
4	CS374TFD	Information Storage Management
5	CS374TFE	Intelligent Software Defined Networks
6	CS374TFF	Mathematical Morphology and Applications

VII SEMESTER INSTITUTIONAL ELECTIVES- II (GROUP G)

Sl. No.	Course Code	Course Title	Credits
1	AI375TGA	Data and Story Telling	3
2	AS375TGB	Aircraft Systems	3
3	BT375TGC	Health Care Technology for Engineers	3
4	CH375TGD	Green and Hydrogen Technology	3
5	CM375TGE	Chemistry of materials and molecular analysis	3
6	CS375TGF	Prompt Engineering	3
7	CV375TGG	Solid Waste Management and Statutory rules	3
8	CV375TGH	Freight Transportation Systems and Logistics	3
9	EC375TGI	IoT for smart systems	3
10	EE375TGJ	E Mobility	3
11	EI375TGK	Disease and Diagnostics an Engineering Perspective	3
12	ET375TGL	Space Technology and Applications	3
13	IM375TGM	Project Management	3
14	IM375TGN	Global Supply chain management	3
15	MA375TGO	Statistical methods for engineers	3
16	ME375TGP	Industry 5.0	3
17	PY375TGQ	Principles of Astrophysics	3
18	MA375TGR	Mathematics of Music	3
19	HS375TGS	Cognitive Psychology	3
20	HS375TGT	Principles & Practices of Cyber Law	3

Semester: VII						
INDIAN KNOWLEDGE SYSTEM						
Common Course						
Theory						
Course Code	:	HS271TA		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45 Hrs		SEE Duration	:	3 Hours
Unit-I						09 Hrs
Introduction of Indian Knowledge Systems (IKS): Perception and prologue, Historicity of IKS, Indian Traditional Knowledge, Nature and scope, kinds of traditional knowledge, Transition from traditional to western knowledge. The IKS Corpus, Vedic Corpus, Indian Philosophical Systems.						
Unit – II						09 Hrs
Foundational Concepts of Mathematics and Science & Technology: Linguistics; Pānini’s work on Sanskrit Grammar, Phonetics, Ashtadhyayi’s vyakarana. Number System and Units of Measurement: concept of zero, Piṅgala and the Binary system. Knowledge: Framework & Classifications: Prameya, Pramana, Samsaya.						
Unit –III						09 Hrs
Science, Engineering & Technology in IKS: Mathematics; Arithmetic, geometry, trigonometry and algebra, Chanda Sastra of Pingala. Astronomy: Elements of the Indian Calendar, Āryabhaṭīya and the Siddhāntic Tradition, Panchanga, Astronomical Instruments (Yantras) and Jyotish. Engineering & Technology: Metals & Metal Working; Wootz Steel, Iron and Steel in India, Metals and Metal working Technology, Lost-wax casting of idols and artefacts, Dyes and Painting Technology, The art of making perfumes.						
Unit –IV						09 Hrs
Yoga and Ayurveda: Tridoṣas, Triguṇa System, Body-Mind-Intellect-Consciousness Complex, Sixty-four art forms and occupational skills (64 Kalas). Irrigation Systems and Water management Practices Town Planning & Architecture: Indian Architecture in India, Vāstu Śāstra; Eight limbs of Vāstu, Town Planning; Temple Architecture						
Unit –V						09 Hrs
Governance, Public Administration & Management: Rāmāyaṇa on great attributes, Arthaśāstra – Governance & Administration, Janapada, Durga, Kosa, Danda, Mitra. IKS & United Nations Sustainable development goals. Safeguarding traditional Indian Knowledge.						

Course Outcomes: After completing the course, the students will be able to:-	
CO1	Explain the nature, scope, and historical development of Indian Knowledge Systems and differentiate traditional knowledge from modern scientific paradigms
CO2	Analyze the foundational concepts of Indian contributions to linguistics, mathematics, and science, including works of Pānini, Piṅgala and others
CO3	Evaluate the applications of traditional Indian technologies in areas such as astronomy, metallurgy, architecture, and medicine.
CO4	Assess the relevance of Indian Knowledge Systems in the context of sustainable development goals and propose methods to safeguard and integrate IKS in contemporary society

Reference Books	
1.	Introduction to Indian Knowledge System Concepts & Applications, B Mahadevan, Vinayaka Rajat Bhat, R N Nagendra Pavana, PHI Learning publishers, ISBN-13: 978-9391818203
2.	Traditional Knowledge System in India, Amit Jha, 2009, Atlantic Publishers and Distributors (P) Ltd., ISBN-13:8126912230-978
3.	Knowledge Traditions and Practices of India, Kapil Kapoor, Avadesh Kumar Singh, Vol. 1, 2005, DK Print World (P) Ltd., ISBN 81-246-0334,
Suggested Web Links:	
1.	https://www.youtube.com/watch?v=LZP1StpYEPM
2.	http://nptel.ac.in/courses/121106003/

3.	http://www.iitkgp.ac.in/department/KS;jsessionid=C5042785F727F6EB46CBF432D7683B63 (Centre of Excellence for Indian Knowledge System, IIT Kharagpur)
4.	https://www.wipo.int/pressroom/en/briefs/tk_ip.html
5.	https://unctad.org/system/files/official-document/ditcted10_en.pdf
6.	http://nbaindia.org/uploaded/docs/traditionalknowledge_190707.pdf
7.	https://unfoundation.org/what-we-do/issues/sustainable-development-goals/?gclid=EAIaIQobChMInp-Jtb_p8gIVTeN3Ch27LAmPEAAAYASAAEgIm1vD_BwE

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). Two tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
PARALLEL ARCHITECTURE AND GPU PROGRAMMING (Theory and Practice)						
Course Code	:	CS372IA		CIE	:	100+50 Marks
Credits: L:T:P	:	3:0:1		SEE	:	100+50 Marks
Total Hours	:	45L+30P		SEE Duration	:	03 + 03 Hours

Unit-I		9 Hrs
Fundamentals of computer design: Introduction; Defining computer architecture; Dependability, Measuring, reporting and summarizing Performance attributes; Quantitative Principles of computer design Pipelining: Introduction, pipeline hazards Instruction level parallelism (ILP): ILP basic concepts and challenges, basic compiler techniques for exposing ILP, reducing branch costs with prediction, overcoming data hazards with dynamic scheduling, hardware based speculation.		
Unit – II		9 Hrs
Multiprocessors and Thread level parallelism: Introduction, Symmetric shared memory architectures; Performance of symmetric shared memory multiprocessors, Distributed shared memory and directory-based coherence, Basics of synchronization, Models of memory consistency.		
Unit –III		9 Hrs
Data-Level Parallelism in Vector, SIMD, and GPU Architectures: Introduction, Vector Architecture, SIMD Instruction Set Extensions for Multimedia, Graphics Processing Units, Detecting and Enhancing Loop-Level Parallelism. GPU programming with CUDA : Introduction, CUDA's programming model: threads, blocks, and grids, CUDA's execution model: streaming multiprocessors and warps , CUDA compilation process, Putting together a CUDA project. Memory hierarchy:, Local memory/registers, Shared memory , Constant memory, Texture and surface memory		
Unit –IV		9 Hrs
Introduction to Parallel Programming Principles of Parallel Algorithm design: Preliminaries, Decomposition Techniques, Characteristics of Tasks and Interactions, Mapping Techniques for Load Balancing, Methods for containing Interaction Overheads, Parallel Algorithms Models. Programming Using the Using Message Passing Paradigm: Principles of Message Passing Programming, Building Blocks, MPI, Topologies and Embedding , Overlapping Communication with Computation, Collective Communication and computation operations, Groups and Communicators.		
Unit –V		9 Hrs
Parallel Programming in OpenACC OpenACC Syntax, Compute Constructs, Data environment, Loop level parallelism- Kernels Versus Parallel Loops, Three Levels of Parallelism, Other Loop Constructs, Programming Tools for OpenACC – Common Characteristics of Architectures, Compiling OpenACC Code Applied GPU Programming: Application Case Study using pyTorch, TensorFlow, Python Numba		

Course Outcomes: After completing the course, the students will be able to	
CO1:	Understand core concepts of computer architecture to investigate parallel algorithms for solving computational problems.
CO2:	Analyze the performance of parallel programming models and techniques and develop proficiency in parallel algorithm design using MPI, OpenACC, and CUDA.
CO3:	Design and implement parallel algorithms using appropriate parallel programming models
CO4:	Demonstrate Parallel computing concepts for suitable compute intensive real time applications

Reference Books	
1	John L Hennessy, David A Patterson; “Computer Architecture: A Quantitative Approach”, Elsevier, 6 th Edition; 2017, eBook ISBN: 9780128119068, Paperback ISBN: 9780128119051
2	Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar : Introduction to Parallel Computing, Second Edition Pearson Education, 2013, ISBN 13: 9788131708071
3	Sunita Chandrasekaran, Guido Juckeland, OpenACC for Programmers: Concepts and Strategies, Addison- Wesley; 1st edition (9 May 2018), ISBN-13: 978-0134694283
4	Gerassimos Barlas, “Multicore and GPU Programming An Integrated Approach”, Elsevier, 2nd Edition, 2023, ISBN: 978-0-12-814120-5.
5	https://developer.nvidia.com/how-to-cuda-python

LABORATORY COMPONENT	
PART – A	
Students are supposed to execute the programs on computationally intensive algorithms using OpenMP, MPI, CUDA and OpenACC	
PART – B	
Students are supposed to demonstrate a mini project using any of the parallel programming concepts.	

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). Two tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
4.	LAB: Conduction of laboratory exercises, lab report, observation, and analysis (40 Marks) and lab test (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE 50 MARKS	50
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		150

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
Student is required to perform Computer Simulation/ Develop a prototype or model as the case may be and present the results in the form of a presentation. Further, students have to submit a poster for exhibition and also a report.		
1	Design and development of the project	20
2	Presentation of the working model/simulation results/prototype building	20
3	Viva Voce	10
TOTAL		50

Semester: VII						
COMPUTER GRAPHICS AND VIRTUAL REALITY (Theory and Practice)						
Course Code	:	CS373IA		CIE	:	100+50 Marks
Credits: L:T:P	:	3:0:1		SEE	:	100+50 Marks
Total Hours	:	45L+30P		SEE Duration	:	03 + 03 Hours

Unit-I		9 Hrs
Introduction to Computer Graphics and Virtual Reality: Application areas of Computer Graphics, Introduction to Graphics Programming with OpenGL, The openGL API: Graphics Functions, The Graphics Pipeline and state Machines, The openGL Interface, Primitives and Attributes, Polygon Basics: polygon types in openGL, Attributes, Color, RGB Color, Indexed Color, Control Functions, The Three- Dimensional Sierpinski Gasket. Display Lists Definition and execution of display Lists, Programming. Introduction to Virtual Reality: The three I's of virtual reality, commercial VR technology and the five classic components of a VR system.		
Unit – II		9 Hrs
Input and Output Devices: Input and Interaction: Input Devices. Physical Input Devices, Logical Devices. Measure and trigger. Input Modes. Event-Driven Input: Using the pointing device, Window events, and Keyboard events. Menus. VR related Input Devices: Trackers, Navigation, and Gesture Interfaces; VR related Output Devices: sound displays & haptic feedback		
Unit –III		9 Hrs
Raster graphics algorithms and Geometric Transformations: Points and lines, line drawing algorithms, mid-point circle algorithm; Filled area primitives: Scan line polygon fill algorithm, boundary-fill and flood-fill algorithms. 2-D viewing: The viewing pipeline, viewing coordinate reference frame, window to view-port coordinate transformation, viewing functions, Cohen-Sutherland and Liang Barsky line clipping algorithms, Sutherland –Hodgeman polygon clipping algorithm. Geometric Transformations: 2-D geometrical transformations: Translation, Scaling, Rotation, reflection and shear transformations, matrix representations and homogeneous coordinates, composite transforms, transformations between coordinate systems.		
Unit –IV		9 Hrs
Geometric Transformations (contd..) 3-D Geometrical Transformations: Translation, Scaling, Rotation. Viewing and Visible Surface Detection: Viewing pipeline: viewing coordinates, Aspect Ratio and view ports, view volume, 3-D clipping. Projections: Classification of planar geometric projections, Projections in openGL; Visible surface detection: Classification, back-face detection, depth-buffer, scan-line, depth sorting, BSP-tree methods, area sub-division and octree methods.		
Unit –V		9 Hrs
Modelling in Virtual Reality: Geometric modelling: Virtual Object Shape, Object Visual Appearance, Kinematics Modelling: Homogeneous Transformation Matrices, Object Position, Transformation Invariants, Object Hierarchies, Viewing the Three-Dimensional World, Physical Modelling: Collision Detection, Surface Deformation, Force Computation, Force Smoothing and Mapping, Haptic Texturing, Behaviour modelling		

Course Outcomes: After completing the course, the students will be able to	
CO1:	Apply the basic concepts of Computer Graphics & Virtual Reality which illustrates the use of the pipeline architecture, OpenGL library, VR tools.
CO2:	Analyze and make an appropriate choice of methods required for computer representation of 2D/3D objects and modelling in Virtual Reality.
CO3:	Design Computer Graphics and Virtual Reality applications using OpenGL library & VR tools.
CO4:	Implement common geometric construction & VR techniques and develop Computer Graphics/ VR based solution to Engineering applications / real-world problems.
CO5:	Exhibit team work and effective oral/written communication skills to accomplish a common goal of solving Computer Graphics/Virtual Reality problems with the engineering community and society at large,

Reference Books	
1	Computer Graphics with OpenGL, Donald D. Hearn, M. Pauline Baker, Warren Carithers, 4 th Edition, 2011, Pearson Education, ISBN-13 : 978-0136053583
2	Interactive Computer Graphics: A Top-Down Approach Using OpenGL, Edward Angel, 5 th Edition, 2010, Pearson Education, ISBN: 9780321535863
3	Computer Graphics, Zhigang Xiang and Roy Plastock, 2 nd Edition, 2007, ASIN: 0070601658, Tata McGraw-Hill, ISBN-13: 978-0070601659.
4	Burdea, G. C. and P. Coffet. Virtual Reality Technology, 2 nd Edition. Wiley-IEEE Press, 2003/2006, ISBN-13 : 978-0471360896
5	Alan Craig, William Sherman and Jeffrey Will, Developing Virtual Reality Applications: Foundations of Effective Design, Morgan Kaufmann, 2009, ISBN: 978-0123749437

LABORATORY COMPONENT	
PART-A	
	- Write a program to generate a line using Bresenham's line drawing technique. Consider slopes greater than one and slopes less than one. User must able to draw as many lines and specify inputs through keyboard/mouse. - Write a program to generate a circle using Bresenham's circle drawing. User can specify inputs through keyboard/mouse. - Design and develop an OpenGL application program to create two windows. Draw a rectangle of specified width and height by setting four different colors to its corners in the first window. In the second window, draw the same rectangle and spin it continuously. Use the double buffer concept. - Design and develop C program using OpenGL libraries to create two windows: display a cylinder in one window and parallelepiped in second window. Note: Create a cylinder and a parallelepiped by extruding a circle and quadrilateral respectively. - Write a program to recursively subdivides a tetrahedron to form 3D Sierpinski gasket. The number of recursive steps is to be specified at execution time. - Write a program to demonstrate the approximation of a 3D sphere with appropriate mathematical formulations. Write the complete C program to approximate a sphere using OpenGL primitives. - Write a program to fill any given 2D polygon using Scan-line area filling algorithm. - Write a program to implement the Cohen Sutherland line clipping algorithm. Make provision to specify the input for multiple lines, window for clipping and viewport for displaying the clipped image.

9. Write a program to **implement the Liang-Barsky line clipping algorithm**. Make provision to specify the input for multiple lines, window for clipping and viewport for displaying the clipped image.
10. Write a program to **implement the Cohen-Hodgeman polygon clipping algorithm**. Make provision to specify the input polygon and window for clipping.
11. Write a program to **create a house like figure and perform the following operations**.
 - i. Rotate it about a given fixed point using OpenGL transformation functions.
 - ii. Reflect it about an axis $y=mx+c$ using OpenGL transformation functions.
12. Write a program to **create a color cube and spin it** using OpenGL transformations.
The output must be adjusted suitably when the window is resized or moved to a new position.

PART-B

Practice Programs on Virtual Reality (Use VR Tools like Unity 3D, Ureal Engine, Blender):

1. Create virtual objects (cube, sphere, cylinder) and apply materials to simulate realistic appearance.
2. Simulate object movement using translation, rotation, and scaling via transformation matrices.
3. Simulate collision detection between two objects and apply a reaction (e.g., bounce, stop, change color).

Open-Ended learning is to be demonstrated by Case study of any Virtual Reality tools (development of a small application or mini project using VR tools).

Case study of any Virtual Reality tool and Implementation

- Students to explore the tools which provide a VR or 360-degree experience at the primary, elementary and secondary levels.
- Demonstrate the Case-Study implemented.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). Two tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
4.	LAB: Conduction of laboratory exercises, lab report, observation, and analysis (40 Marks) and lab test (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE 50 MARKS	50
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		150

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
Student is required to perform Computer Simulation/ Develop a prototype or model as the case may be and present the results in the form of a presentation. Further, students have to submit a poster for exhibition and also a report.		
1	Design and development of the project	20
2	Presentation of the working model/simulation results/prototype building	20
3	Viva Voce	10
TOTAL		50

Semester: VII						
DEEP LEARNING (GROUP-F: Professional Core Elective) (Theory)						
Course Code	:	CS374TFA		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	03 Hours

Unit-I	10Hrs
Neural Networks: What is a neural network, Models of a Neuron, Activation functions, Network Architectures, Knowledge representation, Learning Process. Deep Feedforward Networks: Multilayer Perceptron, Example: Learning XOR, Gradient-Based Learning, Hidden Units, Architecture Design, Back-Propagation Algorithm.	
Unit – II	9Hrs
Convolutional Networks: Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior, Variants of the basic convolution function, Structured Outputs, Data types, Efficient Convolution Algorithms, Random or Unsupervised features, The Neuroscientific basis for convolutional networks.	
Unit –III	10Hrs
Sequence Modeling: Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks, Echo State Networks, The Long Short-Term Memory and Other Gated RNNs.	
Unit –IV	9Hrs
Autoencoders: Undercomplete Autoencoders, Regularized Autoencoders, representational Power, Layer Size and Depth, Stochastic Encoders and Decoders, Denoising Autoencoders, Contractive Auto encoders, Applications of Autoencoders	
Unit –V	7Hrs
Pretrained models: Lenet, AlexNet, VGGNet, Densenet, Resnet, Improving Deep Neural Networks-Hyperparameter Tuning, Regularization and Optimization. Data Augmentation techniques. Other Architectures: Generative Adversarial Networks, Reinforcement Learning	

Course Outcomes: After completing the course, the students will be able to	
CO1:	Explore key theoretical concepts like the Universal Approximation Theorem, vanishing / exploding gradients, and optimization methods.
CO2:	Analyse the fundamental concepts of Deep Learning, and its various architecture learning models, including Neural Networks, backpropagation, gradient descent, and different Network Architectures (feedforward, convolutional, recurrent) Learning tasks for various applications.
CO3:	Apply the Deep learning model approaches to know the strengths and weaknesses of the architecture by empirical results. Apply appropriate concepts like Recurrent, Recursive Nets and Auto-encoder models to specific real time projects and analyse the Optimization techniques.
CO4:	Designing and implement a Deep Learning model as part of an experiential learning initiative in teams to solve societal and environmental problems. Ability to fine tune the model parameters to improve performance, explore and understand the ethical implications and societal impact of deploying deep learning systems in real-world scenarios.

Reference Books	
1	Deep Learning (Adaptive Computation and Machine Learning Series), Ian Good Fellow, Yoshua Bengio and Aaron Courville, MIT Press (3 January 2017), ISBN-13: 978-0262035613.
2	Neural Networks and Learning Machines, Simon S. Haykin, 3rd Edition 2010, PHI Learning, ISBN-9789332586253, 933258625X.
3	Introduction to Artificial Neural Networks, Gunjan Goswami, S.K. Kataria & Sons; 2012 Edition, ISBN-13: 978-9350142967.
4	Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms, Nikhil Buduma, by O'Reilly Publications, 2016 Edition, ISBN-13: 978-1491925614.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). Two tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY)		100

RUBRIC FOR THE SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII					
CYBER SECURITY FOR INDUSTRY 4.0 (GROUP-F: Professional Core Elective) (Theory) (Common to CS & IS)					
Course Code	:	CS374TFB		CIE	: 100 Marks
Credits: L:T:P	:	3:0:0		SEE	: 100 Marks
Total Hours	:	45L		SEE Duration	: 03 Hours

Unit-I	8 Hrs
Introduction: Defining Cyberspace and Cybersecurity, The Value of Standards and Best Practices Documents, The Standard of Good Practice for Information Security, The ISO/IEC 27000 Suite of Information Security Standards, NIST Cybersecurity Framework and Security Documents, The CIS Critical Security Controls for Effective Cyber Defense, COBIT 5 for Information Security, Payment Card Industry Data Security Standard (PCI DSS), ITU-T Security Documents, Effective Cybersecurity.	
Unit – II	9 Hrs
Industrial Network Protocols: Overview of Industrial network protocols, Modbus, ICCP/TASE3, DNP3, OLE for process control, other Industrial Network Protocols, AMI and Smart Grids. How do Industrial Networks operate: Control System Assets – IEDS, RTUs, PLCs, and HMIs, Network architecture, Control System architecture, Control process management, smart grid operations. Vulnerability and Risk Management: Basic Hacking technique, Accessing Industrial networks – The business networks, the SCADA DMZ, Control Systems, Common Vulnerabilities, Smart grids. Determining vulnerabilities, Vulnerability Management.	
Unit –III	10 Hrs
Addressing Cybersecurity in the Age of Industry 4.0: How is Industry 4.0 being leveraged? Understanding cybersecurity challenges in the age of Industry 4.0, Enumerating the factors influencing IoT/OT security, how to overcome security challenges. Delving into Network Segmentation-Based Reference Architecture: The Purdue Model: Zero-trust architecture, Network segmentation in the IoT/OT environment, Understanding the layers of the Purdue model, how layers disrupt security when not managed well, Data diodes, Data diodes in action in OT/IoT. Case study on Automotive Security: Secure dynamic nonlinear heterogeneous vehicle Platooning: Denial of service cyber-attack case.	
Unit –IV	8 Hrs
Cloud and Edge Computing Security in Industry 4.0: Cloud-Based Industrial Systems, Security concerns in cloud-based industrial systems, overview of Edge Computing, Edge Platforms, use cases for edge computing, Edge to cloud protocols, Edge Computing Security Challenges, Security risks associated with distributed edge devices, Encryption and secure data transmission in hybrid cloud-edge environments.	
Unit –V	10 Hrs
AI and Machine Learning for Industrial Cybersecurity: Introduction to AI and ML in cybersecurity, AI applications in real-time threat detection and mitigation, Adversarial Attacks on AI Models in Manufacturing: Risks and vulnerabilities of AI models in industrial applications, Techniques for attacking and fooling AI-driven security systems, Safeguarding AI models from adversarial attacks, AI for Predictive Maintenance and Anomaly Detection, AI and Block chain for future cybersecurity applications and case studies- Architecture and challenges.	

Course Outcomes: After completing the course, the students will be able to	
CO1:	Comprehend and apply fundamental cybersecurity concepts, frameworks, and standards.
CO2:	Analyze industrial communication protocols such as Modbus, DNP3, and IEC61850.
CO3:	Evaluate and tackle cybersecurity challenges in Industry 4.0
CO4:	Investigate cloud and edge computing models in industrial systems
CO5:	Implement AI and Machine Learning techniques in industrial cybersecurity

Reference Books	
1	Stallings, William. <i>Effective Cybersecurity: A Guide to Using Best Practices and Standards</i> . Addison-Wesley Professional, 2018.
2	Song, H., Rawat, D. B., Jeschke, S., & Brecher, C. (2017). <i>Security in cyber-physical systems: Foundations and applications</i> . Hoboken, NJ: Wiley. ISBN: 978-1-119-22604-8
3	Knapp, Eric D., and Joel Thomas Langill. <i>Industrial Network Security: Securing Critical Infrastructure Networks for Smart Grid, SCADA, and Other Industrial Control Systems</i> . 2nd ed., Syngress, 2014. ISBN-13: 978-0124201149.
4	Jain, Smita, and Vasantha Lakshmi. <i>IoT and OT Security Handbook: Assess Risks, Manage Vulnerabilities, and Monitor Threats with Microsoft Defender for IoT</i> . Packt Publishing, 2023. ISBN-13: 978-1804619803.
5	Lea, Perry. <i>IoT and Edge Computing for Architects: Implementing Edge and IoT Systems from Sensors to Clouds with Communication Systems, Analytics, and Security</i> . 2nd ed., Packt Publishing, 2020. ISBN: 9781839218873.
6	Romdhani, Imed, Loai Tawalbeh, Mamoun Alazab, Yassine Maleh, and Youssef Baddi. <i>Artificial Intelligence and Blockchain for Future Cybersecurity Applications</i> . Springer International Publishing, 2021. ISBN: 9783030745752.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar / presentation / demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100



RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
APPLICATION DELIVERY CONTROLLER AND VIRTUALIZATION (GROUP-F: Professional Core Elective) (Theory)						
Course Code	:	CS374TFC		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	03 Hours

Unit-I	9 Hrs
Load balancers: Concepts of L4 load balancing, Managing application delivery using load balancers, L7 Load balancing, Persistence methods, Health monitoring Application Delivery Controller (ADC): Introduction, Significance of ADC, ADC as compared to legacy load balancer, Overview of ADC use cases	
Unit – II	9 Hrs
SSL protocol details, SSL offloading and acceleration, Deployment models for Enterprise Apps, Deep Packet Inspection (DPI), Web Application Firewalls (WAF), Intrusion Prevention System (IPS), IPS as compared to WAF, Deployment modes for NSX.	
Unit –III	9 Hrs
Traffic Management: Core principles of traffic management, Multiprotocol Label Switching, DNS and global server load balancing, Content switching, AppQoE, TCP and SSL profiles, Introduction to Optimization and Security- use cases.	
Unit –IV	9 Hrs
Virtualization and Cloud: Essentials of virtualization and cloud computing, Cloud computing infrastructure, Framework Architecture of Sensor—Cloud Integration, Virtual Machine Migration, Virtual Machine provision process, Virtual Machine Migration Services, Overview of Public clouds like AWS, Azure & Google cloud, Virtualized Data Center Use Case, Importance of virtualizing ADCs	
Unit –V	9 Hrs
Micro services and Containers : Introduction to Micro services & Containers, Dockers, Container Orchestration, Kubernetes, Overview of k8s and k3s, Monitoring, Logging & Tracing tools, Integration with Platforms like Splunk, Elastic Search, Kafka.	

Course Outcomes: After completing the course, the students will be able to	
CO1:	Examine the relevance of various network functionalities like load balancer, ADC, WAF and SSL offloading.
CO2:	Identify the components of application delivery controller and its importance in network traffic management and security
CO3:	Analyze the performance issues of internetwork and explore protocols related to cloud, traffic and global server management
CO4:	Investigate the relevance of virtualization and cloud native architectures in evolving business scenario.

Reference Books

1	Rick Roetenberg, Marius Sandbu, "Mastering NetScaler VPX," 2 nd edition, Packt Publishing, ISBN: 978-1-78528-898-2, 2015
2	Citrix ADC 13.0, Citrix Product Documentation from Citrix Website
3	Citrix NetScaler Deployment Guide and Citrix whitepapers from Citrix website
4	Rajkumar Buyya, James Broberg, Andrzej Goscinski, "CLOUD COMPUTING Principles and Paradigms," Wiley Publications, ISBN 978-0-470-88799-8, 2011
5	Deepak Vohra, "Kubernetes Microservices with Docker," ISBN-13: 978-1-4842-1906-5, 2016
6	Instructor notes

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)

#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100

RUBRIC FOR THE SEMESTER END EXAMINATION (THEORY)

Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII					
INFROMATION STORAGE MANAGEMENT (GROUP-F: Professional Core Elective) (Theory)					
Course Code	:	CS374TFD		CIE	: 100 Marks
Credits: L:T:P	:	3:0:0		SEE	: 100 Marks
Total Hours	:	40L		SEE Duration	: 03 Hours

Unit-I	8 Hrs
Introduction: Storage System Environment: Information Storage, Evolution of Storage Technology and Architecture, Data Center Infrastructure, Key Challenges in Managing Information, Information Lifecycle Components of Storage System Environment, Disk Drive Components, Disk Drive Performance, Fundamental Laws Governing Disk Performance, Logical Components of the Host, Application Requirements and Disk Performance, Storage provisioning (including virtual provisioning)	
Unit – II	8Hrs
Data Protection, Intelligent Storage system: Implementation of RAID, RAID Array Components, RAID Levels, RAID Comparison, RAID Impact on Disk Performance, Hot Spares Components of an Intelligent Storage System, Intelligent Storage array object-based storage and Unified Storage platform	
Unit –III	8Hrs
Direct-Attached Storage, SCSI, and Storage Area Networks: Types of DAS, DAS Benefits and Limitations, Disk Drive Interfaces, Introduction to Parallel SCSI, Overview of Fibre Channel, The SAN and Its Evolution, Components of SAN, FC Connectivity, Fibre Channel Ports, Fibre Channel Architecture, Zoning, Fibre Channel Login Types, FC Topologies Three-site remote replication	
Unit –IV	8Hrs
Backup Purpose, Backup Considerations, Backup Granularity, Recovery Considerations, Backup Methods, Backup Process, Backup and restore Operations, Backup Topologies, Backup in NAS Environments, Backup Technologies. Local Replication, Remote Replication: Source and Target, Uses of Local Replicas, Data Consistency, Local Replication Technologies, Restore and Restart Considerations, Creating Multiple Replicas, Management Interface, Modes of Remote Replication, Remote Replication Technologies, Network Infrastructure, Continuous data protection.	
Unit –V	8Hrs
Securing the Storage Infrastructure, Managing the Storage Infrastructure: Storage Security Framework, Risk Triad, Storage Security Domains, Security Implementations in Storage Networking Monitoring the Storage Infrastructure, Storage Management Activities, Storage Infrastructure Management Challenges, Developing an Ideal Solution, Cloud Infrastructure for storage.	

Course Outcomes: After completing the course, the students will be able to	
CO1:	Analyze various storage architectures along with data center infrastructure requirements.
CO2:	Identify physical and logical components of a storage infrastructure with RAID levels data protection mechanism.
CO3:	Evaluate Storage Area Network types with backup and replications technologies.
CO4:	Monitor and Manage Storage Infrastructure with Security Implementations.

Reference Books	
1	Information storage and management- Somasundaram, Gnanasundaram, Alok Shrivatsava, 2nd Edition, 2015, Wiley publishing ISBN 978-81-265-3750-1.
2	Storage Networking-Real World Skills for the CompTIA Storage+ Certification and Beyond by Nigel Poulton, Publishers, SYBEX a Wiley brand, 2015: ISBN-13 : 978-8126557677
3	Storage Networks Explained – by Ulf Troppens, Wolfgang Muller-Freidt, Rainer Wolafka, IBM Storage Software Development, Germany. Publishers: Wiley
4	Storage Networks: The Complete Reference – Robert Spalding, 2003, Tata McGraw Hill, ISBN: 978-007224764
5	Introduction to Storage Area Networks - Jon Tate, Pall Beck, Hector Hugo, Ibarra Shanmuganathan Kumaravel, Libor Miklas, 9th Edition , December 2017, IBM Redbooks, ISBN-13: 9780738442884.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100

RUBRIC FOR THE SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
INTELLIGENT SOFTWARE DEFINED NETWORKS (GROUP-F: Professional Core Elective) (Theory)						
Course Code	:	CS374TFE		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	03 Hours

Unit-I	09 Hrs
The Genesis of SDN: The Evolution of Networking Technology, Forerunners of SDN, Software Defined Networking is Born, Sustaining SDN Interoperability, Legacy Mechanisms Evolve Toward SDN, Network Virtualization. How SDN Works: Fundamental Characteristics of SDN, SDN Operation, SDN Devices, SDN Controller, SDN Applications, Alternate SDN Methods	
Unit – II	09 Hrs
The OpenFlow Specification - OpenFlow Overview, OpenFlow 1.0 and OpenFlow Basics, OpenFlow 1.1 Additions, OpenFlow 1.2 Additions, OpenFlow 1.3 Additions, OpenFlow Limitations.	
Unit –III	09 Hrs
Alternative Definition of SDN: Potential drawbacks of Open SDN, SDN via APIs, SDN via Hypervisor based overlays, SDN via Opening up the Device. Network function virtualization. Alternative overlap and rakning.	
Unit –IV	09 Hrs
SDN in the Data Center- Data Center Definition, Data Center Demands, Tunneling Technologies for the Data Center, Path Technologies in the Data Center, Ethernet Fabrics in the Data Center, SDN Use Cases in the Data Center, Open SDN versus Overlays in the Data Center, Real-World Data Center Implementations. SDN in Other Environments - Consistent Policy Configuration, Global Network View, Wide Area Networks, Service Provider and Carrier Networks, Campus Networks, Hospitality Networks, Mobile Networks.	
Unit –V	09 Hrs
Intelligent Software Defined Network: Artificial intelligence enabled software[1]defined networking: a comprehensive overview, Network AI: An Intelligent Network Architecture for Self-Learning Control Strategies in Software Defined Networks, Intelligent Routing based on Reinforcement Learning for Software-Defined Networking	

Course Outcomes: After completing the course, the students will be able to	
CO1:	Understand the fundamental definitions, standards and protocols for Software defined Networks (SDN)
CO2:	Explore network programmability through different components such as network programming switches and controller that develop into SDN framework
CO3:	Design network programmable applications using SDN frameworks
CO4:	Analyze the applicability of SDN for future network programmability.

Reference Books

1	Software Defined Networks: A Comprehensive Approach, by Paul Goransson and Chuck Black, Morgan Kaufmann, June 2014, Print Book ISBN: 9780124166752, eBook ISBN : 9780124166844
2	SDN: Software Defined Networks, An Authoritative Review of Network Programmability Technologies, By Thomas D. Nadeau, Ken Gray Publisher: O'Reilly Media, August 2013, ISBN: 978-1-4493-4230-2, ISBN 10:1-4493-4230-2.
3	Network Innovation through OpenFlow and SDN: Principles and Design, Edited by Fei Hu, CRC Press, ISBN-10: 1466572094, 2014.
4	Software defined networks: Design and Deployment, Patricia A. Morreale and James M. Anderson. CRC Press, First edition, December 2014, ISBN: 9781482238631
5	Latah, Majd, and Levent Toker. "Artificial intelligence enabled software-defined networking: a comprehensive overview." IET networks 8.2 (2019): 79-99. (UNIT 5)
6	Yao, Haipeng, et al. "NetworkAI: An intelligent network architecture for self-learning control strategies in software defined networks." IEEE Internet of Things Journal 5.6 (2018): 4319-4327. (UNIT 5)
7	Casas-Velasco, Daniela M., Oscar Mauricio Caicedo Rendon, and Nelson LS da Fonseca. "Intelligent routing based on reinforcement learning for software-defined networking." IEEE Transactions on Network and Service Management 18.1 (2020): 870-881. (UNIT 5)

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)

#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100



RUBRIC FOR THE SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
MATHEMATICAL MORPHOLOGY AND APPLICATIONS (GROUP-F: Professional Core Elective) (Theory)						
Course Code	:	CS374TFF		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	03 Hours

Unit-I	9 Hrs
Introduction to mathematical morphology: Minkowski addition and subtraction, Structuring element and its decompositions. Fundamental morphological operators: Erosion, Dilation, Opening, Closing. Binary Vs Greyscale morphological operations.	
Unit – II	9 Hrs
Morphological reconstructions: Hit-or-Miss transformation, Skeletonization, Coding of binary image via skeletonization, Morphological shape decomposition, Morphological thinning, thickening, pruning. Granulometry, classification, texture analysis: Binary and greyscale granulometries, pattern spectra analysis.	
Unit –III	9 Hrs
Morphological Filtering and Segmentation: Multiscale morphological transformations, Top-Hat and Bottom-Hat transformations, Alternative Sequential filtering, Segmentation.	
Unit –IV	9 Hrs
Geodesic transformations and metrics: Geodesic morphology, Graph-based morphology, City-Block metric, Chess board metric, Euclidean metric, Geodesic distance, Dilation distance, Hausdorff dilation and erosion distances. Efficient implementation of morphological operators.	
Unit –V	9 Hrs
Applications in GISci: Grayscale Granulometric Analysis, Morphological Convexity Measures for Terrestrial Basins Derived from Digital Elevation Models, Logistic Map- A Toy Model, Logistic Equation in Modeling the Geomorphological Phenomena (Lakes)	

Course Outcomes: After completing the course, the students will be able to	
CO1:	Examine morphological principles, methods, efficient algorithms for their computation;
CO2:	Analyze morphological properties and understand the theoretical limits
CO3:	Apply morphological techniques to solve problems in image analysis and GISci
CO4:	Evaluate the performance of different morphological approaches and choose the most suitable method for a given task

Reference Books	
1	J.Serra, 1988, “Image Analysis and Mathematical Morphology: Theoretical Advances”, Academic Press, p. 411.
2	L. Najman and H. Talbot (Eds.), 2010, “Mathematical Morphology”, Wiley, p. 50.
3	B.S.Daya Sagar, “Mathematical Morphology in Geomorphology and GISci,” ISBN 9781138374591, Published by Chapman & Hall , September 18, 2018
4	P. Soille, 2003, Morphological Image Analysis, Principles and Applications, 2 nd edition, Berlin: Springer Verlag.
5	Jean Paul Serra, Jean Serra, “Image Analysis and Mathematical Morphology, Academic Press”, 1982 ; Original from, Cornell University
6	Laurent Najman, Junior Barrera, B. S. Daya Sagar, Petros Maragos, and Dan Schonfeld (Eds.), Special Issue on 'Filtering and Segmentation with Mathematical Morphology' IEEE Journal of Selected Topics in Signal Processing, v. 6, no. 7, p. 737-886, 2012.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100

RUBRIC FOR THE SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII					
DATA AND STORY TELLING					
Institutional Elective – II (Group G)					
(Theory)					
Course Code	:	AI375TGA	CIE	:	100 Marks
Credits: L:T:P	:	3:0:0	SEE	:	100 Marks
Total Hours	:	45L	SEE Duration	:	3Hours
Unit-I					09 Hrs
Understanding Data: Types, Sources, and Preparation: What is data and why it matters, Types of data – qualitative, quantitative, structured, unstructured, 3Vs of big data – volume, velocity, variety, Common data sources – government portals, sensors, APIs, crowd sourced platforms, Data representations – tabular, relational, JSON, XML formats, Introduction to spreadsheets and data frames, Basics of data preprocessing – missing value handling, normalization, type casting, cleaning workflows using Python (pandas) or spreadsheets					
Unit – II					09 Hrs
Data Collection for Exploratory Research: Methods of primary data collection – surveys, interviews, FGDs, observations, Designing structured questionnaires and interview guides, Ensuring question clarity, neutrality, and goal alignment, Sampling strategies – random, stratified, purposive sampling, Field planning and respondent communication, Ethical considerations – consent, privacy, data sensitivity, Transcription and textual data cleaning.					
Unit –III					09 Hrs
Sensor Data Collection and logging: Introduction to IoT and embedded data logging, Sensor interfacing – temperature, humidity, motion, light sensors, Analog-to-digital conversion and signal conditioning, Sampling rates and real-time clock synchronization, Wireless transmission – BLE, Wi-Fi, LoRa, Local and remote data logging – SD cards, cloud platforms, Handling data quality issues – noise, missing values, power constraints, security and privacy (Man in the middle attacks). Real-time plotting, Visualizing trends, anomalies, and multi-sensor overlays, Debugging and optimizing embedded data pipelines, Mapping Urban Tree Inequality: A Data-Driven Look at Green Access, Building a Local Data Chain to Map Green Equity in Our Cities, Project Pakshi					
Unit –IV					09 Hrs
Storytelling through Data: Communicating patterns and trends with clarity, Building simple, reproducible analysis notebooks, Introduction to data storytelling and presentation, Structuring analysis reports – insights, visuals, limitations, Creating visual narratives – dashboards, summaries, infographics, Ethical data interpretation – avoiding bias and misrepresentation					
Unit –V					09 Hrs
Statistics Done Wrong: Statistical Significance, Statistical Power, and the Problem of Underpowered Studies, Pseudo replication, Model Abuse, Researcher Freedom, Hiding the data. Final project – design, collect, visualize, and present data using field instruments, embedded systems, or secondary sources					

Reference Books	
1.	The Craft of Research, Wayne C. Booth, Gregory G. Colomb, and Joseph M. Williams, University of Chicago Press, Fifth Edition, 2024, ISBN-13: 978-0226826677
2.	Sensor Data Analysis and Management: The Role of Deep Learning, Editors: A. Suresh, R. Udendhran, M. S. Irfan Ahmed, Wiley-IEEE Press, 2021, ISBN-13: 978-1119682424
3.	Storytelling with Data: A Data Visualization Guide for Business Professionals, Cole Nussbaumer Knaflic, Wiley Publications, 2015, ISBN-13: 978-1119002253
4.	Statistics Done Wrong: The Woefully Complete Guide, Alex Reinhart, No Starch Press, 2015, ISBN-13: 978-1593276201

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
AIRCRAFT SYSTEMS INSTITUTIONAL ELECTIVE – II (GROUP G) (Theory)						
Course Code	:	AS375TGB		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	3.00 Hours
Unit-I						09 Hrs
Flight Control Systems: History of Airplanes, Anatomy of an aircraft, Anatomy of an aircraft, Primary and secondary flight control surfaces, Flight control linkage system, Conventional Systems, Power assisted and fully powered flight controls, electronic flight control system-fly by wire system.						
Unit – II						09 Hrs
Aircraft Hydraulic & Pneumatic Systems: Components of a typical Hydraulic system, Working or hydraulic system, Power packs, Hydraulic actuators, Pressure regulating devices, Hydraulic pumps, Pneumatic system and components, Use of bleed air, Landing gear and braking,						
Unit –III						09 Hrs
Aircraft Fuel Systems: APU, RAT, Characteristics of aircraft fuel system, Fuel system and its components, different types of fuel tanks, types of fuel lines, Fuel pumps-classification, Fuel control unit, Gravity feed and pressure feed fuel systems,						
Unit –IV						09 Hrs
Environmental Control Systems: Air-conditioning system, vapour cycle system, deicing and anti-icing system, Fire detection- warning and suppression. Crew escape aids. Engine Systems: Engine starting sequence, Starting and Ignition systems, Engine oils and a typical lubricating system.						
Unit –V						09 Hrs
Aircraft Instruments: Instruments displays, Instrumentation grouping, Navigation instruments, Radio instruments, Hydraulic and Engine instruments. Air Data Instruments: Basic air data system and probes, Mach meter, Air speed indicator, Vertical speed indicator, Barometric pressure sensing, Altimeter, Air data alerting system- angle of attack sensing, stall warning, Mach warning, altitude alerting system.						

Course Outcomes: After completing the course, the students will be able to:	
CO1	Demonstrate a comprehensive understanding of the fundamental components and operational principles of major aircraft systems
CO2	Analyze and explain the functions of various subsystems of aircraft in emphasizing their roles in safe and efficient flight operations
CO3	Apply knowledge of aircraft systems to identify typical issues and malfunctions related to flight controls and other subsystems
CO4	Evaluate the the regulatory considerations in aircraft system technologies for different flight envelopes

Reference Books	
1	Introduction to Flight, John D. Anderson, 7 th Edition, 2011, McGraw-Hill Education, ISBN 9780071086059.
2	Fundamentals Of Flight Vol 4: Aircraft Systems, Lalit Gupta, Op Sharma, Himalayan Books, ISBN-13: 9788170020974
3	Flight stability and automatic control, Nelson R.C, 2nd Edition, 1998, McGraw-Hill International Editions, ISBN 9780071158381.

4	Aircraft Systems: Mechanical, Electrical and Avionics Subsystems Integration, Ian Moir, Allan Seabridge, 3rd Edition, 2008, John Wiley & Sons, ISBN 978111965006.	
RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks adding up to 20 Marks. THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY)		100
RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
HEALTH CARE TECHNOLOGY FOR ENGINEERS						
INSTITUTIONAL ELECTIVE – II (GROUP G)						
(Theory)						
Course Code	:	BT375TGC		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45Hrs		SEE Duration	:	3Hours
Unit-I						09 Hrs
Introduction to Healthcare Technology: National Digital Health Mission & Healthcare, IT Infrastructure, Telemedicine and remote healthcare applications, Basic Terminology and Role of Engineers in Healthcare Technology; Definition and importance of healthcare technology, Roles and responsibilities of engineers in the healthcare sector, Interdisciplinary collaboration between healthcare professionals and engineers, Key technologies used in modern healthcare systems.						
Unit – II						09 Hrs
Healthcare Data Management and Digital Epidemiology: Introduction to healthcare data types (structured, unstructured, real-time data), EHR in healthcare and its applications, Data security and privacy in healthcare, Role of digital epidemiology in tracking and predicting diseases, Use of AI and machine learning in healthcare data analysis.						
Unit –III						09 Hrs
Technology-Led Healthcare: Clinical Decision Support Systems (CDSS) and AI in Diagnosis Machine Learning and Deep Learning in Healthcare, Patient Monitoring Systems (PMS) and IoT Applications, Wearable Devices for Health Tracking. Technology-Led Healthcare: Quality Management Systems in Healthcare, Infection Prevention and Waste Management Technologies, Medical Device Development and Quality Assurance, Smart Hospitals: Case Study of a Technology-Enabled Hospital.						
Unit –IV						09 Hrs
Regulatory, Policy and Cybersecurity Aspects: Legal, Ethical and Intellectual Property Rights in Healthcare Technology, Cybersecurity in Healthcare; Data Protection and Privacy, Health Technology Assessment and Procurement Strategies.						
Unit –V						09 Hrs
Innovations in Healthcare Technology: Simulation and Organ Modelling for Medical Applications,3D Printing and Tissue Engineering in Healthcare, Augmented Reality (AR) and Virtual Reality (VR) in Medical Applications, Robotics and Biomechanics in Healthcare, Emerging Digital Health Technologies and Future Trends.						

Course Outcomes: After completing the course, the students will be able to: -	
CO1:	Design, and implement healthcare technologies effectively
CO2:	Analyze the perspective of sensing and imaging in technology for the better health
CO3:	Ensuring compliance with regulatory and cybersecurity frameworks of healthcare data.
CO4:	Evaluate various tools, techniques and advances for better formulation and productivity

Reference Books	
1.	“Biomedical Engineering: Bridging Medicine and Technology.” Saltzman, W. Mark., Cambridge University Press, 2015. https://doi.org/10.1017/CBO9781139583831 .
2.	“Medical Instrumentation: Application and Design.”, Webster, John G., 4th ed., John Wiley & Sons, 2010.
3.	“Artificial Intelligence in Healthcare”. Mahajan, Parag., 2nd (General) ed., Updated 2022.
4.	“Healthcare Information Technology Exam Guide for CHTS and CAHIMS Certifications”. McCormick, Karen A., Barbara Gugerty, and James E. Mattison., McGraw Hill Professional, 2017.
5.	“Introduction to Biomedical Engineering”. Enderle, John D., and Joseph D. Bronzino, editors., 3rd ed., Academic Press, 2012. ISBN: 978-0-12-374979-6.
6.	“Healthcare Information Privacy and Security: Regulatory Compliance and Data Security in the Age of Electronic Health Records”. Robichau, Bernard Peter, Apress, 2014.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only; wherein one sub division will be a caselet in the related topics)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 4 : Question 9 or 10	16
TOTAL		100

Semester: VII						
GREEN AND HYDROGEN TECHNOLOGY						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	CH375TGD		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	3 Hours
Unit-I						09 Hrs
Hydrogen: Peculiarity and Types						
Salient features of hydrogen, properties of hydrogen, terminology and types of hydrogen, advantages, disadvantages, comparison with other fuels, and global status of supply and demand.						
Unit – II						09 Hrs
Hydrogen Generation						
Generation of different types of hydrogen, conventional methods (generation from non-renewable sources), nonconventional methods (generation from renewable sources).						
Unit –III						09 Hrs
Hydrogen Storage						
Storage as compressed gas, storage as cryogenic liquid, storage as metal hydrides, storage through liquid organic hydrogen carriers, and other storage methods.						
Unit –IV						09 Hrs
Hydrogen Handling and Safety						
Classification of hydrogen hazards, compressed and liquid hydrogen related hazards, regulation, codes and standards related to hydrogen handling and transport.						
Unit –V						09 Hrs
Hydrogen Applications						
Applications of hydrogen in various sectors such as refineries, petrochemicals, fertilizer industries, transport and automotive sectors, steel industries, and AI in hydrogen technology.						

Course Outcomes	
CO1	Understand the importance of hydrogen and its use as an energy carrier
CO2	Explain the production, storage and handling of hydrogen
CO3	Analyze the need for hydrogen as an alternate fuel and the associated challenges
CO4	Appraise the importance of safety, regulations and codes

Reference Books	
1.	Hydrogen Safety: Production, Transport, Storage, Use, and the Environment, Fotis Rigas, CRC Press, Taylor & Francis Group, 2 nd Edition, 2023, ISBN: 9781003313007
2.	Hydrogen Fuel: Production, Transport and Storage, Gupta, R. B., CRC Press, Taylor & Francis Group, 1 st Edition, 2009, ISBN: 9780429147364
2.	Hydrogen Production: Electrolysis, Agata Godula-Jopek, Wiley-VCH, 1 st Edition, 2015, ISBN:9783527333424
3.	Handbook of Hydrogen Storage, Michael Hirscher, Wiley-VCH, 1 st Edition, 2010, ISBN:9783527322732
4.	Fuel Cell Systems Explained, James Larminie and Andrew Dicks, John Wiley & Sons, 2 nd Edition, 2003, ISBN 978 0470 848579

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
CHEMISTRY OF MATERIALS AND MOLECULAR ANALYSIS						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	CM375TGE		CIE Marks	:	100 Marks
Credits L: T: P	:	3:0:0		SEE Marks	:	100 Marks
Total Hrs	:	45L		SEE duration	:	3Hours
UNIT – I					09 Hrs	
Advanced Materials for Smart Packaging: Biodegradable polymers, Bio-polymers, Nanomaterials, and Active packaging materials-Self-healing films, antimicrobial coatings and moisture-sensitive barriers for food and pharmaceutical industries. Thermal-responsive packaging used in food safety, drug stability and electronics protection.						
UNIT – II					09 Hrs	
Innovations in Computational Chemistry: Introduction to ChemDraw and its Interface- basic to advanced functionalities for chemical drawing, reaction representation, and integration with computational tools. Cheminformatics and molecular screening. -molecular modeling and simulation techniques- molecular docking, virtual screening, and quantitative structure-activity relationships (QSAR)- generative AI for novel drug molecules- neural networks for toxicity prediction and AI-assisted biomolecular structure prediction.						
UNIT – III					09 Hrs	
IR Spectroscopy: Introduction, principle, molecular vibrations, vibrational frequency, number of fundamental vibrations, factors influencing fundamental vibrations, instrumentation of IR spectrophotometer, application of IR spectroscopy in characterization of functional groups- IR spectroscopy in forensic science, pharmaceuticals, environmental monitoring and material characterization.						
UNIT – IV					09 Hrs	
UV- visible Spectroscopy: Introduction-Electronic transitions- factors influencing position and intensity of absorption bands-absorption spectra of dienes, polyene and α , β -unsaturated carbonyl compounds, Bandgap calculations utilizing UV. Working of UV-Vis spectrophotometer. AI-driven spectral analysis, and computational approaches for UV-Vis data interpretation. UV-Vis spectroscopy in nanomaterials and biomedical applications.						
UNIT – V					09 Hrs	
NMR spectroscopy: H^1 NMR Spectroscopy: Basic concepts- relaxation process. NMR spectrometer-FT NMR- Solvents used in NMR, internal standards-chemical equivalence - chemical shift-Factors affecting chemical shifts-shielding and deshielding effects – chemical and magnetic equivalent –magnetic anisotropy-spin-spin splitting rules- Application of NMR on in characterization of compounds- magnetic resonance imaging (MRI)-Bio-Engineering- Bio-Imaging. Problems on prediction of structure of compounds. Basics of Solid-State NMR.						

Course Outcomes: After completing the course, the students will be able to	
CO1:	Apply the principles of chemistry for the synthesis of smart packaging materials, cheminformatics, drug discovery and materials analysis.
CO2:	Utilize the knowledge of chemistry, AI-driven molecular modelling and spectral analysis to identify compounds and predict toxicity.
CO3:	Analyse IR, UV-Vis, and NMR spectroscopic data for material characterization and biomolecular studies.
CO4:	Propose smart packaging solutions and AI-assisted molecular screening methods for biomedical and pharmaceutical applications

Reference Books	
1.	<i>“Introduction to Spectroscopy”</i> by Donald L. Pavia, Gary M. Lampman, George S. Kriz, and James A. Vyvyan, Published by Cengage Learning, 5th edition, ISBN- 978-1285460123
2.	“Computational Chemistry: Introduction to the Theory and Applications of Molecular and Quantum Mechanics” – Errol G. Lewars. Published by Springer 2024, 4th edition, ISBN- 978-3031514425
3.	<i>“Handbook of Materials Structures, Properties, Processing and Performance”</i> by Lawrence E. Murr Published by Springer, ISBN- 978-3319018140.
4.	<i>“Handbook of Biodegradable Polymers-Applications in Biomedical Sciences, Industry, and the Environment”</i> by <u>Shakeel Ahmed</u> and <u>Riyaz Ali Osmani</u> Published by Jenny Stanford Publishing 2024, ISBN -9789814968843
5.	<i>“An Introduction to Chemoinformatics”</i> by Andrew R. Leach and Valerie J. Gillet, published by Springer ISBN 978-1402062902.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
PROMPT ENGINEERING						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	CS375TGF		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	42 L		SEE Duration	:	03 Hours
Unit-I						8 Hrs
Introduction to Prompt Engineering: Raise of Context Learning, Prompts, Prompt Engineering, LLM Settings, Basic Prompt/ Basics of prompting, Elements of a Prompt, Settings for Prompting Language Model/ General Tips for Designing Prompts, Designing Prompts for Different Tasks: few examples of common tasks using different prompts Text Summarization, Information Extraction, Question Answering, Text Classification, Conversation/Role Playing, Code Generation, Reasoning						
Unit – II						8 Hrs
Techniques for Effective Prompts: Techniques designed to improve performance on complex tasks - Zero-Shot Prompting, Few-shot prompting, Chain-of-thought (CoT) prompting, Zero-Shot CoT, Self-Consistency, Knowledge Generation Prompting, Program-aided Language Model (PAL), React, Directional Stimulus Prompting Prompt Tuning vs. Fine-Tuning: Introduction and Types for Tuning, Tools & Libraries for Tuning- Hugging Face PEFT library (Parameter-Efficient Fine-Tuning), Trade-off decisions: when to prompt-tune vs. fine-tune						
Unit –III						8 Hrs
Best Practices in Prompt Engineering Tools & IDEs: Capabilities include: Developing and experimenting with prompts, Evaluating prompts. Versioning and deploying prompts; Advanced prompting techniques: advanced applications with LLMs LLMs and external tools/APIs -- LLMs with External Tools; Data-augmented Generation – Steps, External Data, QA with sources, Summarization using sources Prompt Debugging & Interpretability Tools: Importance of interpretability in LLMs- Visualization tools- LIT (Language Interpretability Tool by Google) - OpenAI Playground, debug tools						
Unit –IV						8 Hrs
Applications of Prompt Engineering: LLM Applications: Function Calling with LLMs - Getting Started with Function Calling, Function Calling with GPT-4, Function Calling with Open-Source LLMs. Function Calling Use Cases: Conversational Agents, Natural Language Understanding, Math Problem Solving, API Integration, Information Extraction						
Unit –V						8 Hrs
Ethics in AI-Generated Content: Introduction to AI Ethics in Generation, Key Ethical Concerns in Prompting- Bias and Discrimination, Misinformation and Hallucination. Opportunities: Model safety, Prompt Injection, Prompt Leaking, Jail Breaking; Reinforcement Learning from Human Feedback (RLHF)						

Course Outcomes: After completing the course, the students will be able to	
CO1	Demonstrate an understanding of prompt engineering principles including how prompt structure and phrasing impact the performance of AI models.
CO2	Design and implement effective prompts- to create and apply prompts for various natural language processing (NLP) tasks, such as text generation, summarization, and translation, using AI models.
CO3	Critically evaluate the effectiveness of prompts - assess the quality and performance of prompts in terms of accuracy, coherence, and relevance, identifying areas for improvement.
CO4	Apply prompt engineering techniques in real-world scenarios - use prompt engineering strategies to address practical problems in domains such as education, healthcare, and business, demonstrating the applicability of AI-driven solutions.

Reference Books	
1	Unlocking the Secrets of Prompt Engineering, Gilbert Mizrahi, Jan 2024, 1st Edition, Packt Publishing, ISBN-13:978-1835083833
2	Prompt Engineering for Generative AI, James Phoenix, Mike Taylor, May 2024, O'Reilly Media, Inc. ISBN: 9781098153434
3	Prompt Engineering for LLMs, John Berryman, Albert Ziegler, O'Reilly Media, Inc. January 2025, ISBN: 9781098156152
4	OpenAI Cookbook - https://github.com/openai/openai-cookbook “Ethics of Artificial Intelligence and Robotics” – Stanford Encyclopedia of Philosophy- https://plato.stanford.edu/entries/ethics-ai/
5	“The Art of Prompt Engineering with OpenAI API” by Nathan Hunter (Beginner-friendly, focused on real-world use and debugging) ISBN: 9798389421630

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding up to 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (10) Real time problem solving (10) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100

RUBRIC FOR THE SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII					
SOLID WASTE MANAGEMENT AND STATUTORY RULES					
Institutional Elective – II (Group G)					
(Theory)					
Course Code		CV375TGG		CIE	: 100 Marks
Credits: L:T:P	:	3:0:0		SEE	: 100 Marks
Total Hours	:	42L		SEE Duration	: 3 Hours
Unit-I					10 Hrs
Introduction: Present solid waste disposal methods. Merits and demerits of open dumping, incineration, pyrolysis, composting, sanitary landfill, Biogas Scope and importance of scientific solid waste management. Definition and functional elements of solid waste management. Sources: Sources of Solid waste, types of solid waste, composition of municipal solid waste, generation rate, Numerical Problems. Collection and transportation of municipal solid waste: Collection of solid waste- services and systems, Municipal Solid waste (Management and Handling) 2016 rules with amendments. Draft SWM 2024 rules. Site visit to collection system.					
Unit – II					08 Hrs
Composting Aerobic and anaerobic composting - process description, process microbiology, Vermicomposting, Site visit to compost plant, Numerical problems. Sanitary land filling: Definition, advantages and disadvantages, site selection, methods, reaction occurring in landfill- Gas and Leachate movement, Control of gas and leachate movement, Site visit to landfill site.					
Unit –III					08 Hrs
Hazardous waste management: Definitions, Identification of hazardous waste, Classification of hazardous waste, onsite storage, collection, transfer and transport, processing, disposal, Hazardous and other wastes (Management and Transboundary Movement) Rules, 2016 with amendments. Hazardous and other wastes (Management and Transboundary Movement) amendment rules, 2024. Site visit to hazardous landfill site					
Unit –IV					08 Hrs
Bio medical waste management: Classification of bio medical waste, collection, transportation, disposal of bio medical waste, Biomedical waste management (Management & Handling Rules) 2016 with amendments. Site visit to hospital to observe biomedical waste collection and transportation system and visit to biomedical waste incineration plant.					
Unit –V					08 Hrs
E-waste management: Definition, Components, Materials used in manufacturing electronic goods, Recycling and recovery integrated approach. E-waste (Management) Rules 2022 and amendments. Site visit to e- waste treatment plant. Plastic waste management: Manufacturing of plastic with norms. Plastic waste management. Plastic waste management rules 2016 with amendments.					

Course Outcomes: After Completing the course, student will be able to,	
CO1	Understand the current solid waste management system and statutory rules.
CO2	Analyze drawbacks in the present system and provide recycling and disposal options for each type of waste in compliance to rules.
CO3	Distinguish Hazardous waste, Biomedical waste, E waste and to provide scientific management system.
CO4	Evaluate and monitor the Biomedical waste, Hazardous waste, E waste, Plastic and Municipal waste management as per the rules laid by Ministry of Environment, Forest and Climate change.

Reference Books

1.	Integrated Solid Waste Management, George.C.Tchobanoglous, International edition 2022, McGraw hill publication. ISBN 978-0070632370
2.	Electronic waste management, R.E. Hester, Roy M Harrison, , Cambridge, UK, 2009, RSC Publication, ISBN 9780854041121
3.	Solid Waste Management Rules 2016, Ministry of Environment, Forest and Climate Change Notification, New Delhi, 8th April 2016
4.	Hazardous and other wastes (Management and Transboundary Movement) Rules, 2016, Ministry of Environment, Forest and Climate Change Notification, New Delhi, 04th April, 2016.
5.	Biomedical waste management (Management & Handling Rules) 2016, Ministry of Environment & Forest Notification, New Delhi, amendment on 28th March, 2016.
6.	E-waste (Management) Rules 2022 and amendments, Ministry of Environment, Forest and Climate Change Notification, New Delhi, 2 nd November, 2022.
7.	Plastic waste management rules 2016 with amendments., Ministry of Environment, Forest and Climate Change Notification, New Delhi, 6 th July 2022.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)

#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)

Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
FREIGHT TRANSPORTATION SYSTEMS AND LOGISTICS						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	CV375TGH		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	42L		SEE Duration	:	3 Hours
Unit-I						08 Hrs
Characteristics of Freight Transport: Freight Characteristics, Factors influencing Freight Travel, operators, problems in freight transportation, regional and urban goods travel, intermodal freight travel issues, passenger and freight demand models.						
Unit – II						09 Hrs
Freight Demand Estimation: Operations, Planning - purpose, process, Data, Freight Agents, costs, Planning Models and Methods-freight demand estimation and forecasting at the regional and urban level, Freight Generation and Freight Trip Generation, Trend and time series models, freight trip rate models, IO model						
Unit –III						08 Hrs
Freight Transport Planning and Operations: Freight supply – capacity issues; freight productivity and performance; distribution of freight flows; production/consumption to origin/destination, competing modes for specific commodity choice, route planning, scheduling, collection storage, distribution centres, regulation, and enforcement of freight transport.						
Unit –IV						09 Hrs
Logistics and Planning Strategies: Context of Logistics- Activities of Logistics, Aims of Logistics, Importance of Logistics, Current Trends in Logistics; Logistics Strategy- Strategic Decisions, Logistics Strategy, designing a Logistics Strategy; Locating Facilities- Importance of Location, Choosing the Geographic Region, Infinite Set Approaches, Feasible Set Approaches, Network Models, Location Planning; Planning Resources- Types of Planning, Capacity Planning, Adjusting Capacity, Tactical Planning, Schedules						
Unit –V						08 Hrs
Emerging Trends and Case Studies in Transportation Logistics: Sustainable and Green Logistics, Digital Transformation in Logistics-IoT, big data analytics, AI, and blockchain in logistics management.						
Case Studies and Real-World Applications: Case studies on logistics and transportation from various sectors. Analysis of successful logistics systems and lessons learned.						

Course Outcomes	
CO1	Explain the characteristics, components, and current trends in freight transportation and logistics
CO2	Analyze and apply freight demand estimation methods and logistics planning strategies for urban and regional transport systems
CO3	Evaluate freight transport systems and logistics networks with respect to performance, capacity, sustainability, and regulation
CO4	Design integrated freight and logistics solutions using data-driven models, technology, and strategic planning approaches

Reference Books	
1	M. Ben-Akiva, H. Meersman, and E. V. de Voorde, "Freight Transport Modelling" Bingley, U.K.: Emerald Group Publishing, ISBN-13: 9781781902851,2013.
2	P. K. Sarkar, V. Maitri, and G. J. Joshi, "Transportation Planning: Principles, Practices and Policies", 3rd ed. New Delhi, India: PHI Learning, ISBN : 9788195161188, 2024
3	L. Tavasszy and G. de Jong, "Modelling Freight Transport". 1st ed. Amsterdam, Netherlands: Elsevier, ISBN-13: 9780124104006,2013.
4	M. Al-Azzawi, "Freight and Logistics Transport Modelling and Planning: Mathematical Theories and Practical Applications for the Analysis and Forecasting of Freight Transport Systems". Saarbrücken, Germany: LAP Lambert Academic Publishing, ISBN-13: 9783848428380,2012

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
IoT FOR SMART SYSTEMS						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	EC375TGI		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	3 Hours
Unit-I						09 Hrs
Introduction to IoT and Networking Basics						
IoT Architecture Layers (Perception, Network, Application), Overview of Networking Basics: TCP/IP, OSI Model, Wireless Concepts, Device Addressing: IPv4, IPv6, 6LoWPAN, IoT orchestration primitives, IoT Open-Source platforms.						
Unit – II						09 Hrs
IoT Communication Models and Protocols						
Smart Manufacturing-OPC-UA interfacing, Deterministic Ethernet & Time Sensitive Network, PLC gateways, MQTT-Sparkplug B, Predictive maintenance dashboard design with AI-based Optimization.						
Smart Construction Site: MQTT with QoS for equipment-telemetry and asset tracking, Edge gateway (rugged fog node) with aggregating sensor feeds (On-site construction).						
Unit –III						09 Hrs
IoT Networking Technologies						
Smart Healthcare: IEEE 802.15.6 BLE Body Area Networks, NFC for patient data access and device pairing, HL7 and FHIR standards for health data exchange.						
Smart Mobility: Next-gen V2X, LTE-M for roadside units and environmental sensors, Edge-cloud split for traffic-signal optimization, LPWAN & NB-IoT with link-budget exercise.						
Unit –IV						09 Hrs
IoT Software Stack						
Smart Agriculture Systems:						
Device Layer: Sensor nodes run on MCUs with firmware enabling periodic sensing, sleep scheduling, and uplink messaging over LPWAN. Network Layer: Embedded Operating Systems for IoT (Free RTOS). Middleware Layer: Gateway running rule engine and device management, Device abstraction and Service Enablement, Secure bootloaders for wireless firmware updates in remote farms. Application Layer: Web/mobile dashboard for farmers with real-time alerts and analytics, Things Board for drag-and-drop rule engine.						
Unit –V						09 Hrs
Interoperability and Standardization in Practice: Challenges of integrating multi-vendor IoT devices, Interoperability issues in AI-Enabled IoT Deployments, Data Semantics & Model Portability, AI Bias & Fairness Across Domains, Security of AI pipelines, GDPR compliance & data retention policies in IoT platforms.						
Course Outcomes						
CO1	Develop a conceptual framework of IoT architectural layers and connectivity models.					
CO2	Analyze IoT-specific communication models, protocols, and networking technologies.					
CO3	Apply AI techniques for IoT communication optimization and network intelligence.					
CO4	Evaluate secure, energy-efficient IoT networks with intelligent features.					

Reference Books	
1.	Internet of Things: Principles and Paradigms, Rajkumar Buyya, Amir Vahid Dastjerdi, Morgan Kaufmann Publishers, 2nd Edition, 2024, ISBN: 978-0-323-99167-2
2.	Architecting the Internet of Things, Dieter Uckelmann, Mark Harrison, Florian Michahelles Springer, 2nd Edition, 2023 ISBN: 978-3-031-10318-4
3.	IoT Applications for Next-Generation Smart Systems, D. Giusto et al., Springer, 2024, ISBN: 978-1-799-87541-3
4.	Artificial Intelligence for the Internet of Everything, William Lawless, Ranjeev Mittu, Donald Sofge Academic Press, 2023 ISBN: 978-0-12-820600-1

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
E-MOBILITY						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	EE375TGJ		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	3 Hours
Unit-I						09 Hrs
History, Basics of Electric Vehicles, Components of Electric Vehicle, General Layout of EV, EV classification: Battery Electric Vehicles (BEVs), Hybrid Electric Vehicle (HEV), Fuel-Cell Electric Vehicles (FCEVs) Comparison with Internal Combustion Engine: Technology, Advantages & Disadvantages of EV, National Policy for adoption of EVs.						
Unit – II						09 Hrs
Electric Drive-Trains: Introduction to various electric drive-train topologies in EV and HEV, Power flow control in electric drive-train topologies, classification of electric machines used in automobile drivetrains. E-Motor Drives Configuration (Control Block diagrams): Induction Motor Drive, Permanent Magnet (PM) motor Drive & Switched Reluctance Motor (SRM) Drive.						
Unit –III						09 Hrs
Battery Energy Storage: Types of Battery, Introduction to Electrochemical Battery, Electrochemical Reactions, Battery Parameters: Battery Capacity, Discharge Rate, Charging Rate, SOC, SOD, SOH, DOD, Specific Energy, Specific Power, Energy Efficiency. Battery Management Systems (BMS): Introduction to BMS, Objectives of the BMS: Discharging control, Charging control, Cell Balancing; BMS topologies: Distributed Topology, Modular Topology and Centralized Topology.						
Unit –IV						09 Hrs
Energy Storage: Introduction to Energy Storage Requirements in Hybrid and Electric Vehicles, Battery based energy storage, Fuel Cell based energy storage, Super Capacitor based energy storage, Hybridization of different energy storage devices. Introduction to BMS and its topologies. Energy Management Strategies: Introduction to energy management strategies used in hybrid and electric vehicles, Classification of different energy management strategies, Comparison of different energy management strategies and implementation issues of energy management strategies.						
Unit –V						09 Hrs
Charging Infrastructure: Basic Requirements for Charging System, Charger Architectures, Grid Voltages, Frequencies, and Wiring, Charging Standards and Technologies, SAE J1772. On-board chargers and Off-board chargers, Topologies and Standards, Types of Charging Station Charging Station Placement for Electric Vehicles: A Case Study.						

Course Outcomes	
CO1	Analyze the basics of electric and hybrid electric vehicles, their architecture, technologies and modelling.
CO2	Analyze various electric drives suitable for electric vehicles.
CO3	Discuss and implement different energy storage technologies used for electric vehicles and their management system.
CO4	Analyze various charging methods, requirements, standards and types of charging for EV and HEV.

Reference Books	
1.	Modern Electric Vehicle Technology C.C. Chan and K.T. Chau, 1st Edition, 2001, Oxford university press, ISBN 0198504160.
2.	Battery Management system for large Lithium Battery Packs, Davide Andrea, 1st Edition, 2010, ARTECH HOUSE, ISBN-13 978-1-60807-104-3.
3.	Electric Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles, John G. Hayes, G. Abas Goodarzi, 1st Edition, 2018, Wiley, ISBN 9781119063667.
4.	Hybrid Vehicles from Components to System, F. BADIN, Ed, 1st Edition, 2013, Editions Technip, Paris, ISBN 978-2-7108-0994-4.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
DISEASE AND DIAGNOSTICS AN ENGINEERING PERSPECTIVE						
Institutional Electives -II (Group G)						
(Theory)						
Course Code	:	EI375TGK		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	03 Hrs
Unit-I						09 Hrs
Cardiovascular System (CVS) - Functioning of CVS, Generation of Bio-Potentials of CVS, Common cardiovascular ailments. - Arrhythmias, Hypertension, Coronary vascular Diseases, Diagnostic devices- ECG, ECHO. Coronary Angiogram. Implantable devices: Pacemaker.						
Unit – II						09 Hrs
Respiratory System (RS) - Functioning of RS, Common ailments in RS, Lung volume and capacity, Diagnostic Devices- Spirometer and its recent advances, PPG Signals and its application in RS.						
Unit –III						09 Hrs
Renal System - Functioning of Renal System, Common diseases of Renal System, Diagnostics- Ultrasound and its application in Renal System. Therapeutic- Dialysis and its types. Bio-Markers in Renal System, Emerging dialysis technologies: Wearable artificial kidney.						
Unit –IV						09 Hrs
Central Nervous System (CNS) - Basics of CNS functioning, Common ailments, Diagnostics methods-EEG, Electrode Systems, Role of CT scans and MRI. Brain-Computer Interfaces (BCI): Basic concepts and diagnostic/therapeutic potential.						
Unit –V						09 Hrs
Introduction to Artificial Intelligence (AI) in Healthcare - Overview of AI, Machine Learning, Deep Learning, Need for AI in healthcare, Types of data in healthcare: structured, unstructured., Challenges of AI deployment: Data bias, privacy, and integration in clinical workflow.						

Course Outcomes	
CO1	Understand the basic functioning and common disorders of cardiovascular, respiratory, renal, and nervous systems.
CO2	Understand the working principles and clinical relevance of diagnostic devices such as ECG, EEG, spirometer, ultrasound, and imaging systems.
CO3	Analyze biomedical signals and interpret their significance in the diagnosis and monitoring of diseases.
CO4	Evaluate the potential and limitations of emerging technologies such as wearable devices, Brain-Computer Interfaces, and AI in healthcare applications.

Reference Books	
1.	Handbook of Biomedical Instrumentation, R. S. Khandpur, 3 rd Edition, Reprint 2016, Tata McGraw-Hill, ISBN: 9780070473553.
2.	Biomedical Instrumentation and Measurements, Leslie Cromwell & others, 2 nd Edition, Reprint 2015, ISBN: 9780130771315.
3.	Introduction to Biomedical Equipment Technology, Joseph J. Carr and John M. Brown, 4 th Edition, 2000, Pearson, ISBN: 978-0130104922.
4.	Medical Instrumentation Application and Design, John G. Webster, 4 th Edition, 2010, ISBN 13: 978- 0471-67600-3.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII					
SPACE TECHNOLOGY AND APPLICATIONS					
Institutional Elective (Group-G)					
(Theory)					
Course Code	:	ET375TGL		CIE	: 100 Marks
Credits: L:T:P	:	3:0:0		SEE	: 100 Marks
Total Hours	:	45L		SEE Duration	: 03 Hours
UNIT – I					9 Hours
Orbits and Launching Methods: Kepler's Laws, Orbital Elements and Perturbations, Inclined Orbits Antenna Look Angles, Earth Eclipse of Satellite, Sun Transit Outage. Launch Vehicles: Rocketry, Propellants, Propulsion, Combustion, Solid, Liquid and Cryogenic engines, Control and Guidance system, Ion propulsion and Nuclear Propulsion.					
UNIT – II					9 Hours
Satellite Sub-Systems: Classification of satellites., Payloads, Subsystems-Altitude and orbit control system, TT&C Sub-System, Power Systems. Communication sub systems: Transponder, Satellite antennas and Parameters, Satellite Antennas in Practice					
UNIT – III					8 Hours
Satellite Communications: LEO, MEO and GEO orbits, Multiple Access Techniques. -TDMA, FDMA and CDMA Space applications: Telephony, V-SAT, DBS system, Satellite Radio and TV, Tele-Education, Tele-medicine, Satellite navigation, GPS					
UNIT – IV					8 Hours
Remote Sensing: Remote Sensing – An Overview, Classification of Satellite Remote Sensing Systems, Remote Sensing Satellite Payloads, Sensor Parameters. Image processing techniques, Remote sensing cameras. Applications of Remote Sensing Satellites: - Land Cover Classification, Land Cover Change Detection Water Quality Monitoring and Management, Flood Monitoring, weather modelling, Cyclone predictions, Disaster and flood warning, rainfall predictions using satellites.					
UNIT – V					8 Hours
Space Missions: Technology missions, deep space planetary missions, Lunar missions, zero gravity experiments, space biology and international space Missions. Advanced space systems: planetary payloads, space shuttle, space station, Inter-space communication systems.					

Course Outcomes: After going through this course the student will be able to:		
CO1	:	Explore the different types of satellites, their orbital mechanics and associated subsystems.
CO2	:	Apply the basics of launching vehicles, satellites and sub systems for space applications.
CO3	:	Identify the applications of satellite in the Various areas such as communication, remote sensing, metrology etc.
CO4	:	Analyze the technology trends, satellite missions and advanced space systems.

Reference Books	
1.	Satellite Communications, Dennis Roddy, W. Linwood Jones, Jones Linwood, David G. Long, 5ed, McGraw Hill Education, 2024, ISBN-10: 1265372543
2.	Fundamentals of Satellite Communication, K N Raja Rao, PHI, 2012, ISBN: 9788120324015.
3.	Satellite Communication, Timothy Pratt, John Wiley, 3ed, 2019, ISBN: 978-1-119-48217-8
4.	Remote sensing and applications, B C Panda, VIVA books Pvt. Ltd., 2009, ISBN: 108176496308.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
Sl. No.	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & each Quiz will be evaluated for 10 marks, and Final Quiz marks adding up to 20 marks. THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: Question 2	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
MAXIMUM MARKS FOR THE SEE		100

Semester: VII						
PROJECT MANAGEMENT						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	IM375TGM		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	3 Hours
Unit-I						09 Hrs
Introduction: Project, Project management, relationships among portfolio management, program management, project management, and organizational project management, relationship between project management, operations management and organizational strategy, business value, role of the project manager, project management body of knowledge.						
Generation and Screening of Project Ideas: Generation of ideas, monitoring the environment, corporate appraisal, scouting for project ideas, preliminary screening, project rating index, sources of positive net present value.						
Unit – II						09 Hrs
Project Scope Management: Project scope management, collect requirements define scope, create Work Breakdown Structure, validate scope, control scope.						
Organizational influences & Project life cycle: Organizational influences on project management, project state holders & governance, project team, project life cycle.						
Unit –III						09 Hrs
Project Integration Management: Develop project charter, develop project management plan, direct & manage project work, monitor & control project work, perform integrated change control, close project or phase.						
Project Quality management: Plan quality management, perform quality assurance, control quality.						
Unit –IV						09 Hrs
Project Risk Management: Plan risk management, identify risks, perform qualitative risk analysis, perform quantitative risk analysis, plan risk resources, control risk.						
Project Scheduling: Project implementation scheduling, Effective time management, Different scheduling techniques, Resources allocation method, Project life cycle management concepts. Project life cycle costing.						
Unit –V						09 Hrs
Latest Trends and Open-Source Tools in Project Management: Agile and Hybrid Project Management Approaches, Digital Transformation in Project Management, Sustainability and ESG in Project Management, Overview and Classification of PM Software, Introduction and Demonstration of Key Open-Source Tools.						

Course Outcomes	
CO1	Understand the fundamental concepts of project management and its relationship with organizational strategy, operations management, and business value.
CO2	Apply techniques for generating, screening, and evaluating project ideas, considering factors such as net present value and project rating index.
CO3	Create Work Breakdown Structures (WBS), utilization of PERT/CPM for developing project schedule, alongside requirement collection, scope definition, scope validation, and scope control.
CO4	Develop skills in project integration, quality, risk management, and scheduling, enabling effective project planning, execution, monitoring, and control.

Reference Books	
1.	Project Management Institute, “A Guide to the Project Management Body of Knowledge (PMBOK Guide)”, 5th Edition, 2013, ISBN: 978-1-935589-67-9
2.	Harold Kerzner, Project Management A System approach to Planning Scheduling & Controlling, John Wiley & Sons Inc., 11th Edition, 2013, ISBN 978-1-118-02227-6.
3.	Prasanna Chandra, Project Planning Analysis Selection Financing Implementation & Review, Tata McGraw Hill Publication, 7th Edition, 2010, ISBN 0-07-007793-2.
4.	Rory Burke, “Project Management – Planning and Controlling Techniques”, John Wiley & Sons, 4th Edition, 2004, ISBN: 9812-53-121-1

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
GLOBAL SUPPLY CHAIN MANAGEMENT						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	IM375TGN		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	3 Hours
UNIT-I						09 Hrs
Building a Strategic Frame Work to Analyze Supply Chains: Definition and Objective of Supply Chain, the importance of Supply Chain Decisions, Decision Phases in a Supply Chain, Process View of Supply Chains. Competitive and Supply Chain Strategies, Achieving Strategic fit, Expanding Strategic Scope. Drivers of Supply Chain Performance, Frame work for Structuring Drivers, Facilities, Inventory, Transportation, Information, Sourcing, Pricing, Infrastructure, International Logistics.						
UNIT – II						09Hrs
Designing The Supply Chain Network: The Role of Distribution in the Supply Chains, Factors influencing Distribution Network design, Design Options for a Distribution Network, Online sales and the Distribution network, Distribution Networks in practice. Factors influencing network design decisions, Framework for Network design decisions, The impact of Globalization on Supply Chain networks. Designing And Planning Transportation Networks: The role of transportation in a Supply chain, Modes of transportation and their performance characteristics, Transportation infrastructure and policies, Design options for a transportation network, Trade-offs in transportation design, Tailored transportation, The role of IT in transportation, Problems.						
UNIT –III						09Hrs
Planning and Managing Inventories in a Supply Chain: The Role of Cycle inventory in a Supply Chain, Economies of Scale to Exploit Fixed costs, Managing Multi-echelon Cycle Inventory. The Role of Safety Inventory in a Supply Chain, Determining appropriate level of Safety inventory, Impact of supply Uncertainty on Safety inventory, Impact of aggregation on safety inventory, impact of replenishment policies on safety inventory, Managing Safety Inventory in a Multi-echelon Supply Chain, The Role of IT in inventory management. Problems						
Unit –IV						09Hrs
Sourcing Decisions in A Supply Chain: The role of sourcing in a supply chain, in-house or outsource, Third-and Fourth-party logistics providers, Total cost of Ownership, Supplier Selection-Auctions and Negotiations, Sharing Risk and Reward in the Supply chain, Ethical Sourcing. Pricing and Revenue management in Supply Chain: The role of pricing and revenue management in supply chain, pricing and revenue management for multiple customer segments, pricing and revenue management for perishable assets, pricing and revenue management for seasonal demand and role of IT in pricing and revenue management.						
UNIT –V						09 Hrs
Digital Supply Chain: The role of IT in a supply chain, the supply chain IT framework, the supply chain macro processes, Lack of Supply Chain co-ordination and the Bullwhip effect, managerial levers to achieve coordination, continuous replenishment and vendor-managed inventories, collaborative planning, forecasting and replenishment (CPFR).						

Course Outcomes	
CO1	Understand supply chain concepts, systemic and strategic role of SCM in global competitive environment.
CO2	Evaluate alternative supply and distribution network structures using optimization models.
CO3	Develop optimal sourcing and inventory policies in the supply chain context.
CO4	Select appropriate information technology frameworks for managing supply chain processes.

Reference Books	
1	Supply Chain Management – Strategy, Planning & Operation, Sunil Chopra, Peter Meindl & D V Kalra, 6 th Edition, 2016, Pearson Education Asia; ISBN: 978-0-13-274395-2.
2	Supply Chain Management – Creating Linkages for Faster Business Turnaround, Sarika Kulkarni & Ashok Sharma, 1 st Edition, 2004, TATA Mc Graw Hill, ISBN: 0-07-058135-5
3	Designing & Managing the Supply Chain – Concepts Strategies and Case Studies, David Simchi Levi, Philip Kaminsky, Edith Simchi Levi & Ravi Shankar, 3 rd Edition, 2008, Mc Graw Hill, ISBN: 978- 0-07-066698-6
4	Modelling the Supply Chain, Jeremy F Shapiro, 2 nd Edition, 2009, Cengage Learning, ISBN 0-495-12609-8.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
STATISTICAL METHODS FOR ENGINEERS						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	MA375TGO		CIE	:	100 Marks
Credits: L: T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	3.00 Hours
Unit-I					09 Hrs	
Multiple Linear Regression: Introduction to multiple linear regression model, Least squares estimation of the parameters, Matrix approach to multiple linear regression, Properties of the least squares estimators, Hypothesis tests -Test for significance of regression, Tests on individual regression coefficients, Confidence intervals - Confidence intervals on individual regression coefficients, Confidence interval on the mean response, Polynomial regression models.						
Unit – II					09 Hrs	
Design and Analysis of Single-Factor Experiments: Designing Engineering Experiments, Completely Randomized Single-Factor Experiment, Analysis of Variance (ANOVA), Multiple Comparisons following the ANOVA, Residual analysis and model checking, The random-effects model, Randomized complete block design - design and statistical Analysis Design of Experiments with Several Factors: Introduction, Factorial experiments, Two-factor factorial experiments, one observation per cell, General factorial experiments - 2^k Factorial designs, 2^2 Design, 2^k Design for $k \geq 3$ factors.						
Unit –III					09 Hrs	
Discriminant Analysis and Factor Analysis: Introduction, Linear Discriminant Model, Linear discriminant function, Discriminant analysis, Principal Component, Factor Analysis, Principal Components versus Factor Analysis, Applications. Conjoint Analysis: Introduction, Additive tables, Multiplicative tables, Computing table margins based on an additive model, Applied conjoint analysis.						
Unit –IV					09 Hrs	
Decision Trees and Clustering: Introduction, Basic tree model, Categorical or Quantitative Predictors, Decision Trees, Regression Trees, Bias-Variance Tradeoff, Pruning and Cross-Validation, Loss functions. Cluster Analysis-Introduction, Types of Clustering, Correlations and Distances, Hierarchical Clustering, Partitioning via k-means.						
Unit –V					09 Hrs	
Time Series Analysis: Time series analysis-Trend, Seasonality, Noise decomposition, Autocorrelation, Stationarity, Lag features, Overview of Autoregressive (AR), Moving Average (MA), Autoregressive Integrated Moving Average (ARIMA) models.						

Course Outcomes:	
CO1	Apply fundamental concepts of multiple regression, design of experiments to analyze and interpret data in discriminant analysis, factor analysis, conjoint analysis, and statistical learning methods.
CO2	Analyze and interpret data using regression, experimental design, and multivariate techniques, and apply suitable statistical learning models to real-world problems.
CO3	Estimate model parameters and interpret results for multiple linear regression, ANOVA, discriminant analysis, factor analysis, and conjoint analysis to support data-driven decision-making using statistical learning techniques.
CO4	Implement multiple linear regression, ANOVA, discriminant analysis, factor analysis, conjoint analysis and time series analysis to support decision-making using statistical methods for solving engineering problems.

Reference Books	
1	An Introduction to Multivariate Analysis, T. W. Anderson, 3 rd Edition, 2003, John Wiley & Sons, New Jersey, ISBN: 0-471-36091-0.
2	Douglas C. Montgomery and George C. Runger, “Applied Statistics and Probability for Engineers”, John Wiley & Sons, 7th Edition, 2019, ISBN: 9781119570615.
3	Trevor Hastie, Robert Tibshirani, Jerome Friedman, “The Elements of Statistical Learning: Data Mining, Inference, and Prediction”, Springer Series in Statistics, 2nd Edition, 2009.
4	Richard A Jhonson and Dean W. Wichern, Applied Multivariate Statistical Analysis, 6 th Edition, Pearson Education, ISBN-13: 978-0-13-187715-3.
5	Statistics I, SYSTAT 10.2, ISBN 81-88341-04-5.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
INDUSTRY 5.0						
Institutional Elective (Group -G)						
(Theory)						
Course Code	:	ME375TGP		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	42 Hrs		SEE Duration	:	3 Hours
Unit – I						06 Hrs
Evolution to Industry 5.0: Historical overview of industrial revolutions. Transition from automation-centric to human-centric manufacturing. Key drivers: personalization, sustainability, and resilience. Comparison between Industry 4.0 and Industry 5.0 frameworks.						
Unit -II						08 Hrs
Human-Machine Collaboration: Introduction to collaborative robots (cobots) and their applications. Designing ergonomic and safe human-robot workspaces. Augmented Reality (AR) and Virtual Reality (VR) in training and maintenance. Case studies on successful human-robot collaboration.						
Unit –III						08 Hrs
Advanced Digital Technologies: Digital twins and their role in predictive maintenance and system optimization. Integration of AI and machine learning in manufacturing processes. Cyber-Physical Systems (CPS) and their applications. Edge computing and real-time data analytics.						
Unit –IV						10 Hrs
Sustainable and Resilient Manufacturing: Principles of sustainable manufacturing and circular economy. Life Cycle Assessment (LCA) and its importance in product design. Strategies for building resilient supply chains. Regulatory frameworks and standards for sustainable practices.						
Unit - V						10 Hrs
Implementation and Case Studies: Roadmap for transitioning to Industry 5.0. Challenges and solutions in adopting Industry 5.0 principles. Case studies from automotive, aerospace, and electronics industries. Future trends and research directions in Industry 5.0.						

Course Outcomes: After completing the course, the students will be able to	
CO1	Assess the challenges of implementing Industry 5.0 principles in diverse industrial applications.
CO2	Analyze the strategies for transitioning from Industry 4.0 to Industry 5.0 in human-centric manufacturing systems.
CO3	Evaluate the benefits of integrating collaborative technologies and sustainable practices in smart manufacturing.
CO4	Examine the role of human-machine collaboration and ethical data-driven decision-making in achieving resilient and adaptive production systems.

Reference Books	
1.	Industry 5.0: The Future of the Industrial Economy, Uthayan Elangovan, 2022, CRC Press, 1 st Edition, ISBN: 9781032041278 (Unit I)
2.	Industry 5.0: Concepts and Strategies for Digital Transformation, Kaswan, Rathi, Garza-Reyes, 2025, CRC Press, 1 st Edition, ISBN: 9781032878218 (Unit II, V)
3.	Industry 5.0: Design, Standards, Techniques and Applications for Manufacturing, Leong, 2024, The IET, 1 st Edition, ISBN: 9781837240098 (Unit IV)
4.	Towards Industry 5.0, Durakbasa, Gencyilmaz, 2022, Springer, 1 st Edition, ISBN: 9783031244568 (Unit III, V)
5.	Intelligent Manufacturing: Smart Choice, Sunil Kumar Wadhwa, 2023, ISBN: 9781802279153

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. Each quiz is evaluated for 10 marks adding up to 20 MARKS	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE THEORY		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
PRINCIPLES OF ASTROPHYSICS						
Category: Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	PY375TGQ		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	42L		SEE Duration	:	3 Hours
Unit-I						08 Hrs
The Celestial Sphere: The Geocentric Universe, Retrograde motion, The Copernican revolution, The celestial sphere, Altitude–Azimuth and Equatorial Coordinate System, Precession.						
Celestial Mechanics: Kepler’s Laws of Planetary Motion from Newton’s laws of mechanics, Position in the Orbit.						
Spectrum of Light: Stellar parallax, The magnitude scale, Apparent and Absolute Magnitude, Blackbody radiation and temperature, Spectral lines - Kirchhoff’s Laws, Spectrographs.						
Unit – II						09 Hrs
Special Relativity: Einstein’s postulates, The relativity of simultaneity, Time dilation, Length contraction, Galilean transformations, Lorentz transformations, Velocity transformation, Space-time diagrams and causality. Relativistic momentum and energy.						
General Relativity: The Principle of Equivalence, Gravitation as space-time curvature, (1) perihelion advance of planet Mercury, (2) gravitational red shift and (3) gravitational bending of light (qualitative).						
Unit –III						08 Hrs
Telescopes: Resolution and the Rayleigh Criterion, Aberrations, The Brightness of an Image and focal ratio, Refracting and Reflecting Telescopes, Adaptive Optics, Electronic Detectors - CCD, Radio Telescopes - Spectral Flux Density, Large Apertures and Interferometry, Infrared, Ultraviolet, X-Ray, and Gamma-Ray Astronomy.						
Unit –IV						09 Hrs
Stars: Stellar Spectra, The Harvard Spectral Classification, The Hertzsprung–Russell Diagram, Binary Stars and Stellar Masses - Classification of Binary Stars, Stellar Structure - Internal Equilibrium Conditions, The Sun - The Atmosphere, Solar Activity; Stellar Energy Sources - Nuclear Fusion Reactions, Stellar Evolution - Evolutionary Time Scales, The Contraction of Stars Towards the Main Sequence, The Main Sequence Phase, The Giant Phase, The Final Stages of Evolution - White Dwarfs, Neutron Stars, Black Holes.						
Unit –V						08 Hrs
Galaxies: The Classification of Galaxies, Masses, Galactic Structures, Structural Components of the Milky Way, Dynamics of Galaxies, Systems of Galaxies, The Origin and Evolution of Galaxies. The Interstellar Medium - Interstellar Dust, Interstellar Gas, Cosmic Rays.						
Cosmology: The Olbers Paradox, Hubble’s Law, Cosmic Microwave Background Radiation, History of the Universe, Astrobiology.						

Course Outcomes	
CO1	Apply the principles of celestial mechanics to describe planetary motion, orbital dynamics, and gravitational interactions in astrophysical systems.
CO2	Interpret the foundational concepts of special relativity and analyse the implications of time dilation, length contraction, and relativistic mass in astronomical contexts.
CO3	Compare and evaluate different types of telescopes and their applications in various regions of the electromagnetic spectrum for observational astronomy.
CO4	Analyse the physical properties, life cycles, and classifications of stars using observational data and theoretical models.

Reference Books	
1.	Bradley W. Carroll, Dale A. Ostlie, An Introduction to Modern Astrophysics: United States Edition, Pearson; 1st edition (7 September 2017), ISBN-10: 1108422160
2.	H. Karttunen, P. Kröger, H. Oja, M. Poutanen, K. J. Donner (Eds.), Fundamental Astronomy, 5th Edition, Springer, 2007, ISBN 978-3-540-34143-7
3	Suresh Chandra Mohit Kumar, A Textbook of Astronomy and Astrophysics, Dreamtech Press, 1 Nov 2019, ISBN-10: 9389520908
4.	Padmanabhan, T., Theoretical Astrophysics, Vols.1-3, 2015, Cambridge University Press, ISBN-9780521016278
5.	Modern Cosmology, Academic Press Inc; 2nd edition, 7 July 2020, ISBN-10: 0128159480,
6.	Harwit, M. Astrophysical Concepts, 4th Edition, 2015, Springer-Verlag, ISBN-10 : 1441921990
7.	Shapiro, Stuart L., and Saul A. Teukolsky. Black Holes, White Dwarfs, and Neutron Stars, 1st Edition, 1983, Wiley, ISBN: 9780471873167

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
MATHEMATICS OF MUSIC						
Institutional Elective – II (Group G)						
(Theory)						
Course Code	:	MA375TGR		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	3 Hours
Unit-I						09 Hrs
Fundamentals of Sound and Musical Structure						
Nature of sound: frequency, amplitude, harmonics, waveforms; Musical pitch and logarithmic perception of frequency, Shruti, Swara, Saptak; Musical notation systems.						
Unit – II						09 Hrs
Tuning Systems and Modular Arithmetic in Music						
Pythagorean tuning, Just Intonation, and Equal Temperament, Circle of Fifths and the irrationality of $\log(3/2)$, Modular arithmetic and pitch class sets, Construction of scales: Melakarta Rāga(Carnatic) and Thaata (Hindustani) systems.						
Unit –III						09 Hrs
Rhythm, Tāla, and Combinatorial Structures						
Tāla systems: concepts of mātra, vibhāga, āvartana; Analysis of Carnatic tālas (Adi, Rupaka, Jhampa, etc.); Combinatorics of tāla patterns and variation (korvais, tihāis); Algorithmic composition of rhythmic phrases						
Unit –IV						09 Hrs
Timbre, Music Analysis and Processing						
Timbre and harmonic spectra, Fourier series and transforms in sound decomposition, Sub-harmonics, Limitation of Fourier Analysis: Noise; Introduction to sound synthesis and spectral envelopes.						
Unit –V						09 Hrs
Algorithmic Music and Perception						
Fractals and recursion in music, Algorithmic and generative composition, Mathematical models of musical perception, Psychoacoustics: pitch, consonance/dissonance, critical bands.						

Course Outcomes	
CO1	Analyze musical phenomena using mathematical tools
CO2	Model rhythm, harmony, and timbre using algebraic and computational methods.
CO3	Apply signal processing concepts to sound synthesis and analysis.
CO4	Explore algorithmic and generative approaches to musical composition.

Reference Books	
1	Benson, D. J. (2006). Music: A Mathematical Offering. Cambridge University Press, ISBN: 978-0521619998.
2	Fauvel, J., Flood, R., & Wilson, R. (Eds.). (2003). Music and Mathematics: From Pythagoras to Fractals. Oxford University Press, ISBN: 978-0199298938.
3	Loy, G. (2006). Musimathics: The Mathematical Foundations of Music (Vol. 1 & 2). MIT Press, ISBN: 9780262516556.
4	Sethares, W. A. (2005). <i>Tuning, Timbre, Spectrum, Scale</i> (2nd ed.). Springer, ISBN: 978-1852337971.
5	Radhika Iyer (2018), Elements Of Indian Music: The Melakarta System, ISBN: 1513460854.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII					
COGNITIVE PSYCHOLOGY					
Institutional Elective Course (Group G)					
Theory					
Course Code	:	HS375TGS		CIE	: 100 Marks
Credits: L:T:P	:	3:0:0		SEE	: 100 Marks
Total Hours	:	45 L		SEE Duration	: 3 Hours
Unit-I					09 Hrs
Fundamentals & current trends in cognitive psychology: Definition, Emergence of cognitive psychology, Cognitive development theories and perspectives; Current status and trends in cognitive Psychology. Research methods in cognitive psychology- goals of research. Distinctive research method. Current areas of research in cognitive psychology, (Educational application, marketing and advertisement).					
Unit – II					09 Hrs
Basic cognitive processes: Sensation and Perception: Sensory receptors and Brain, The constancies, pattern recognition, Modularity, Imagery: Characteristics of Imagery, Cognitive maps. Attention and Information processing: Nature and Types, Theories and models of attention. Neuropsychological studies of Attention. Consciousness: – meaning, Modern Theories and Contemporary Research of Consciousness.					
Unit –III					09 Hrs
Reasoning, Creativity and Problem- Solving: Reasoning definition, types, influencing factors. Creativity- definition, steps involved in creative process, obstacles involved in creativity, enhancing techniques of creativity. Meta cognition: Problem solving, steps in problem solving, types, methods, obstacles and aids of problem Solving.					
Unit –IV					09 Hrs
Psycholinguistics: Definition, characteristics of language, theories - Chomsky. Structure of Language (Properties), Stages in Language Development, Neurological Language. Comprehension and Production. Bilingualism, Multilingualism and Learning disability.					
Unit –V					09 Hrs
Cognitive Neuroscience: Definition and emergence of cognitive neuroscience, Scope of Neuroscience, structure and functions of Brain, Brain Plasticity, Intelligence and Neuroscience. Meta-cognitive strategies. Artificial intelligence, Robotics, Models on Information Processing.					

Course Outcomes: After completing the course, the students will be able to: -	
CO1	Describe the basic theories, principles, and concepts of cognitive psychology as they relate to behaviour and mental processes.
CO2	Define learning and compare and contrast the factors that cognitive, behavioral, and Humanistic theorists believe influence the learning process.
CO3	Develop understanding of psychological attributes such as reasoning, problem solving creativity, resulting in their enhancement and apply effective strategies for self-management and self-improvement.
CO4	Apply the theories into their own and others' lives in order to better understand their personalities and experiences.

Reference Books	
1.	Sterberg R.J and Sternberg Karin (2012) Cognitive Psychology 6 th Edition Woods worth Cengage Learning
2.	Psychology-themes and variations, Wayne Weiten, IV edition, Brooks / Cole Publishing Co.
3.	Psychology Robert A. Baron, III edition (1995) Prentice Hall India.
4.	Understanding Psychology Feldman R. S, IV edition, (1996) McGraw Hill India

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII						
PRINCIPLES & PRACTICES OF CYBER LAW						
Institutional Elective Course (Group G)						
Theory						
Course Code	:	HS375TGT		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45 L		SEE Duration	:	3 Hours
Unit-I					08 Hrs	
Introduction - Origin and meaning of Cyberspace; Introduction to Indian Cyber Law, Distinction between Cyber Crime and Conventional Crime, Cyber Criminals and their Objectives, Kinds of Cyber Crime& Cyber Threats, challenges of cybercrimes, Overview of General Laws and Procedures in India. Cyber Jurisdiction -Concept of Jurisdiction, Jurisdiction in Cyberspace, Issues and concerns of Cyberspace Jurisdiction in India, International position of Cyberspace Jurisdiction, Judicial interpretation of Cyberspace Jurisdiction. Activities: Case Studies and Practical Applications						
Unit – II					08 Hrs	
Information Technology Act: A brief overview of Information Technology Act 2000, IT Act 2000 vs. IT Amendment Act 2008, Relevant provisions from Indian Penal Code, Indian Evidence Act, Bankers Book Evidence Act, Reserve Bank of India Act, etc. Electronic Signature and Digital Signature- Meaning & Concept of Relevance of Signature, Handwritten signature vs Digital Signature, Technological Advancement and development of signature, Digital Signature: IT Act, 2000, Cryptography, Public Key and Private Key, Public Key Infrastructure Electronic Signature vs. Digital Signature, E-Commerce under IT Act2000, Issues and challenges of E-Commerce. Activities: Case Studies and Practical Applications						
Unit –III					08 Hrs	
Data Protection and Privacy Concerns in Cyberspace - Need to protect data in cyberspace, Types of data, Legal framework of data protection, Data protection bill -an overview, GDPR, Concept of privacy, Privacy concerns of cyberspace, Constitutional framework of privacy, Judicial interpretation of privacy in India. Data Privacy and Data Security- Defining data, meta-data, big data, non- personal data. Data protection, Data privacy and data security, Data protection regulations of other countries- General Data Protection Regulations (GDPR),2016 Personal Information Protection and Electronic Documents Act (PIPEDA),, social media- data privacy and security issues. Activities: Case Studies and Practical Applications						
Unit –IV					08 Hrs	
IP Protection Issues in Cyberspace Copyright Issues in Cyberspace- Copyright infringement in digital environment. Indian legal protection of copyright in cyberspace. Trademark Issues in Cyberspace -Domain Name Vs Trademark, Domain Name dispute and Related Laws, Different Form of Domain in Cyberspace. Patent Issues in Cyberspace -Legal position on Computer related Patents - Indian Position on Patents. Activities: Case Studies and Practical Applications						
Unit –V					07 Hrs	
Digital Forensics- Computer Forensics, Mobile Forensics, Forensic Tools, Anti-Forensics Cyber Crime & Criminal Justice Agencies -Cyber Crime Cells, Cyber Crime Appellate- Cyber Crime Investigation, Investigation Procedure- FIR -Charge Sheet						

Course Outcomes: After completing the course, the students will be able to: -	
CO1	Understand the importance of professional practice, Law and Ethics in their personal lives and professional careers.
CO2	Build in Depth Knowledge of Information Technology Act and Legal Frame Work of Right to Privacy, Data Security and Data Protection.
CO3	Identify the bone of contentions of cybercrime investigation techniques, evaluate problem-solving strategies, and develop science-based solutions.
CO4	Develop an Understanding of the Relationship Between E-Commerce and Cyberspace.

Reference Books	
1.	Cyber Law by Dr. Pavan Duggal Publisher: LexisNexis, ISBN-10: 8196241070, ISBN-13: 978-8196241070
2.	Introduction to Information Security and Cyber Laws by Surya Prakash Tripathi, Ritendra Goel, Praveen Kumar Shukla ASIN: 9351194736, Publisher: Dream tech Press, ISBN-10: 9789351194736, ISBN-13: .9351194736-978
3.	Cyber Forensics in India: A Legal Perspective by Nishesh Sharma, 1 st Edition, ISBN: 9788131250709.
4.	Cyber Laws, Justice Yatindra Singh, 6 th Edition, Vol. 1, ISBN: 9789351437338

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100

Semester: VII					
SUMMER INTERNSHIP					
Course Code	:	CS376SI		CIE	: 100 Marks
Credits: L:T:P	:	0:0:3		SEE	: 100 Marks
No. of Weeks	:	08		SEE Duration	: 3 Hours
GUIDELINES					
- The duration of the internship shall be for a period of 8 weeks on full time basis after VI semester final exams and before the commencement of VII semester. - The student must submit letters from the industry clearly specifying his / her name and the duration of the internship on the company letter head with authorized signature. - Internship must be related to the field of specialization of the respective UG programme in which the student has enrolled. - Students undergoing internship training are advised to report their progress and submit periodic progress reports to their respective guides. - Students have to present the internship activities carried out to the departmental committee and only upon approval by the committee, the student can proceed to prepare and submit the hard copy of the final internship report. However, interim or periodic reports as required by the industry / organization can be submitted as per the format acceptable to the respective industry / organizations. - The reports shall be printed on A4 size with 1.5 spacing and Times New Roman with font size 12, outer cover of the report (wrapper) has to be Ivory color for UG circuit Programs and Light Blue for Non-Circuit Programs. - The broad format of the internship final report shall be as follows - Cover Page - Certificate from College - Certificate from Industry / Organization - Acknowledgement - Synopsis - Table of Contents - Chapter 1 - Profile of the Organization: Organizational structure, Products, Services, Business Partners, Financials, Manpower, Societal Concerns, Professional Practices, - Chapter 2 - Activities of the Department - Chapter 3 - Tasks Performed: summary of the tasks performed during 8-week period - Chapter 4 – Reflections: Highlight specific technical and soft skills acquired during internship - References & Annexure					
Course Outcomes: After going through the internship the student will be able to: CO1: Apply Engineering and Management principles CO2: Analyze real-time problems and suggest alternate solutions CO3: Communicate effectively and work in teams CO4: Imbibe the practice of professional ethics and need for lifelong learning.					
Scheme of Continuous Internal Evaluation (CIE): The evaluation committee shall consist of Guide, Professor/Associate Professor and Assistant Professor. The committee shall assess the presentation and the progress reports in two reviews. The evaluation criteria shall be as per the rubrics given below:					
Reviews	Activity				Weightage
Review-I	Explanation of the application of engineering knowledge in industries, ability to comprehend the functioning of the organization/ departments.				50 Marks
Review - II	Importance of resource management, environment and sustainability, presentation skills and report writing				50 Marks

Scheme for Semester End Evaluation (SEE):

The SEE examination shall be conducted by an external examiner (domain expert) and an internal examiner. Evaluation shall be done in batches, not exceeding 6 students per batch.

Scheme of Evaluation for SEE	
Particulars	%Marks
Project Synopsis (Initial Writeup)	10%
Project Demo/Presentation	30%
Methodology and Results Discussion	30%
Project Work Report	10%
Viva-voce	20%
Total	100

Semester: VIII					
MAJOR PROJECT					
Course Code	:	CS481P		CIE	: 100 Marks
Credits: L:T:P	:	0:0:12		SEE	: 100 Marks
Hours/Week	:	24		SEE Duration	: 03 Hours
GUIDELINES					
- The project topic, title and synopsis have to be finalized and submitted to their respective internal guide(s) before the beginning of the 8th semester. - The detailed Synopsis (approved by the department Project Review Committee) has to be submitted during the 1st week after the commencement of 8th semester.					
Batch Formation: - Students are free to choose their project partners from within the program or any other program. - Each student in the team must contribute towards the successful completion of the project. - The project may be carried out In-house / Industry / R & D Institution. The project work is to be carried out by a team of two to four students, in exceptional cases where a student is placed in a company and offered an internship through the competitive process or student is selected for internship at national or international level through competitive process, the student can work independently. - The students are allowed to do either a project for full 5 days in the industry or full 5 days in the college. - In case the project work is carried out outside Bengaluru, such students must be available during Project Evaluation process scheduled by the respective departments and they must also interact with their guide regularly through Email / Webinar / Skype etc.					
Project Topic Selection: The topics of the project work must be in the field of respective program areas or in line with CoE's (Centre of Excellence) identified by the college or List of project areas as given by industry/Faculty. The projects as far as possible should have societal relevance with focus on sustainability.					
Project Evaluation: - Continuous monitoring of project work will be carried out and cumulative evaluation will be done. - The students are required to meet their internal guides once in a week to report their progress in project work. - Weekly Activity Report (WAR) has to be maintained in the form of a diary by the project batch and the same has to be discussed with the Internal Guide regularly. - In case of Industry project, during the course of project work, the internal guides will have continuous interaction with external guides and will visit the industry at least twice during the project period. - For CIE assessment the project groups must give a final seminar with the draft copy of the project report. - The presentation by each group will be for 20-30 minutes and every member of the team needs to justify the contributions to the project. - The project team is required to submit Hard copies of the detailed Project Report in the prescribed format to the department. - For CIE 50% weightage should be given to the project guide and 50% weightage to the project evaluation committee. - Before the final evaluations the project group is required to produce a No dues certificate from Industry, Central Library and Department.					
Course Outcomes: After going through the major project the student will be able to: CO1: Apply knowledge of mathematics, science and engineering to solve respective engineering domain problems.					

CO2: Design, develop, present and document innovative/multidisciplinary modules for a complete engineering system.

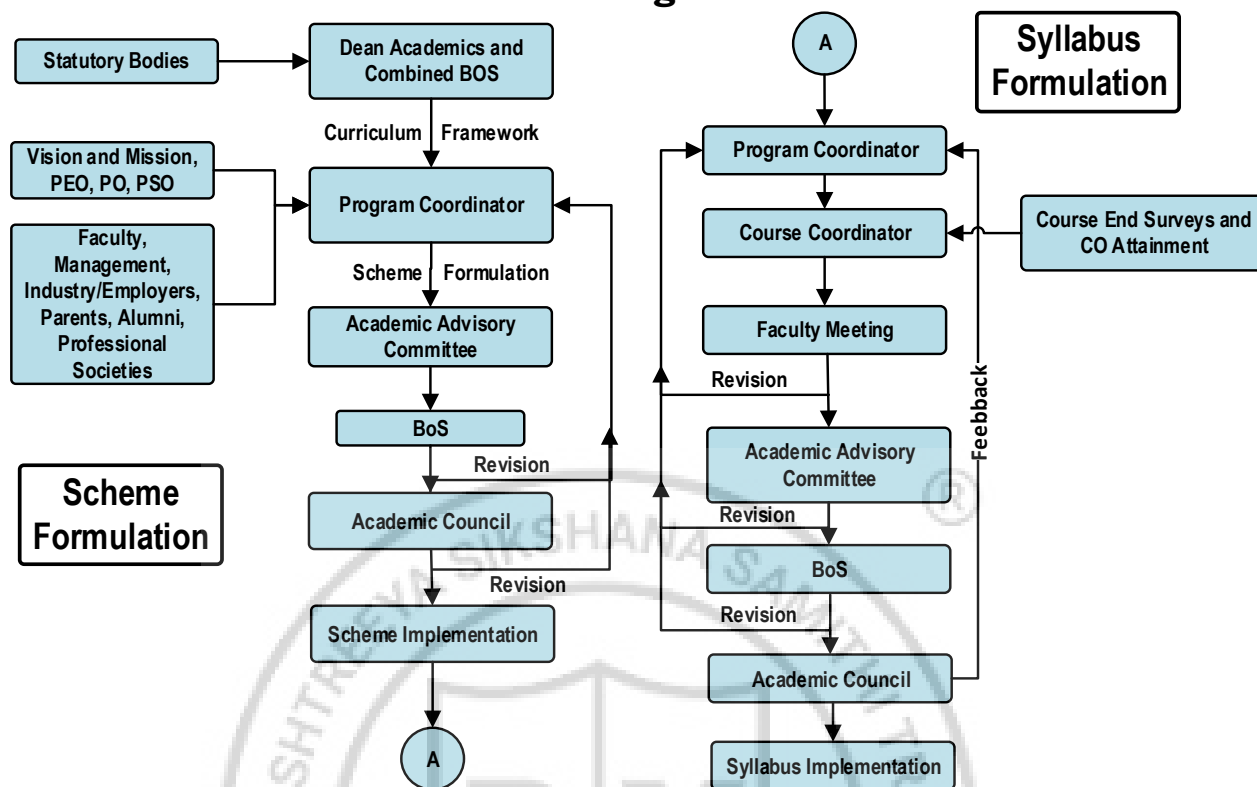
CO3: Use modern engineering tools, software and equipment to solve problem and engage in life-long learning to follow technological developments.

CO4: Function effectively as an individual, or leader in diverse teams, with the understanding of professional ethics and responsibilities.

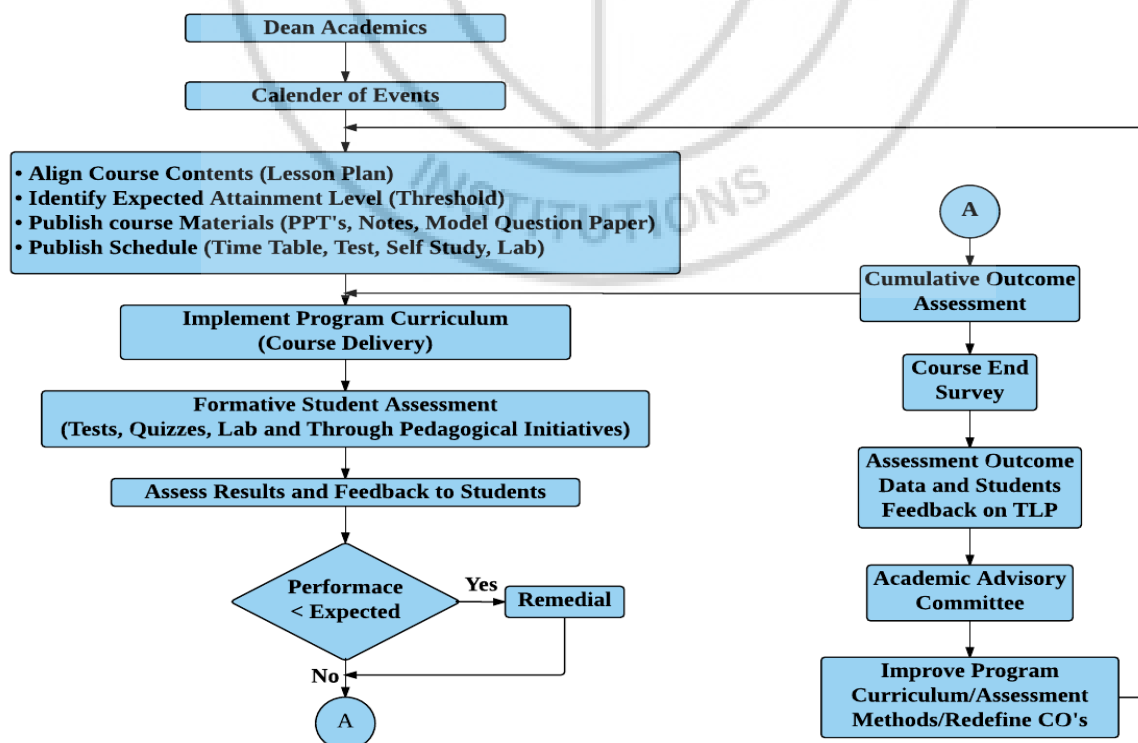
Scheme of Continuous Internal Evaluation (CIE):

Scheme of Evaluation for CIE		Scheme of Evaluation for SEE	
Particulars	%Marks	Particulars	%Marks
Project Evaluation I	10%	Project Synopsis (Initial Writeup)	10%
Project Evaluation II	20%	Project Demo/Presentation	25%
Project Evaluation III	25%	Methodology, Results and Discussion	25%
Project Evaluation Phase-IV (Submission of Draft Project Report for Verification) (Project Final Internal Evaluation)	25%	Project Work Report	15%
Publication	20%	Viva-voce	25%
Total	100	Total	100

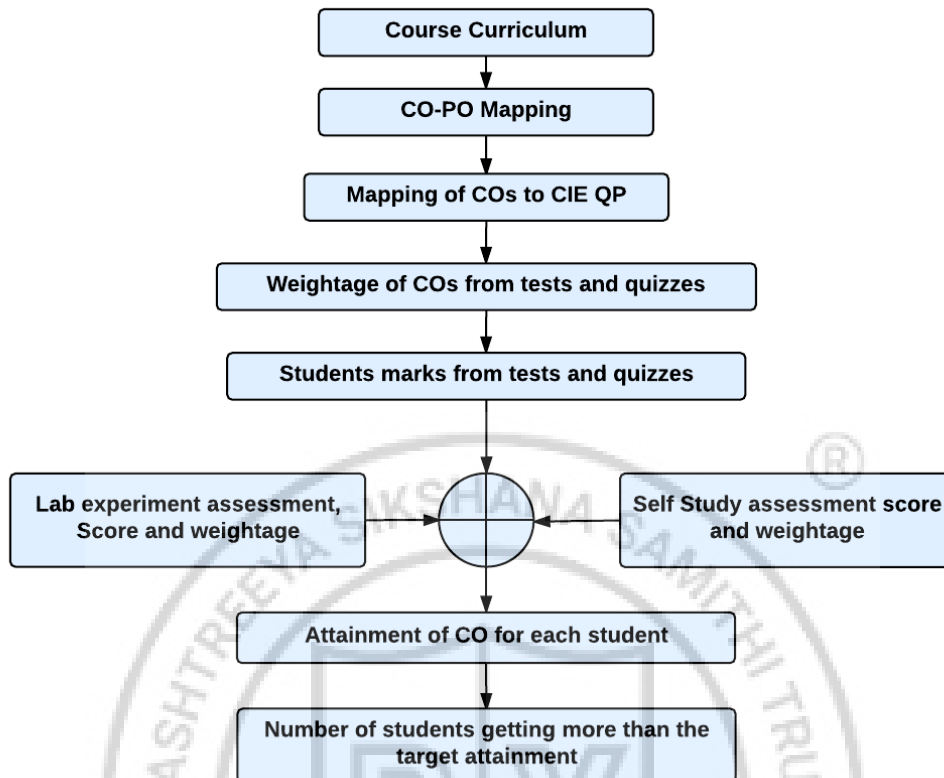
Curriculum Design Process



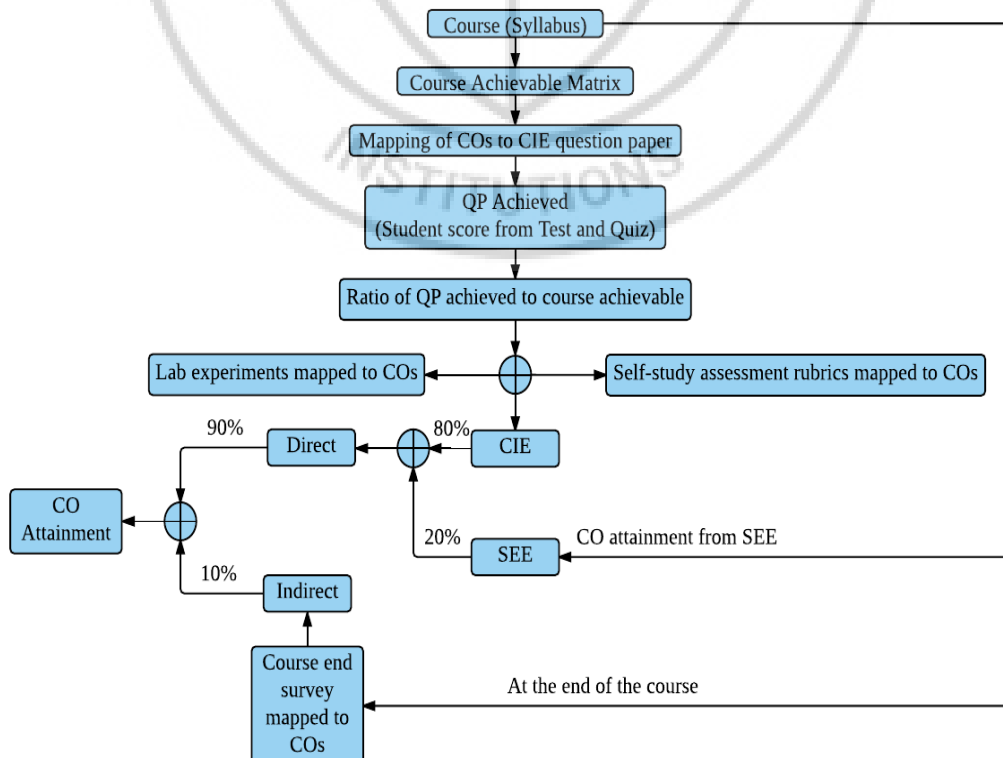
Academic Planning and Implementation



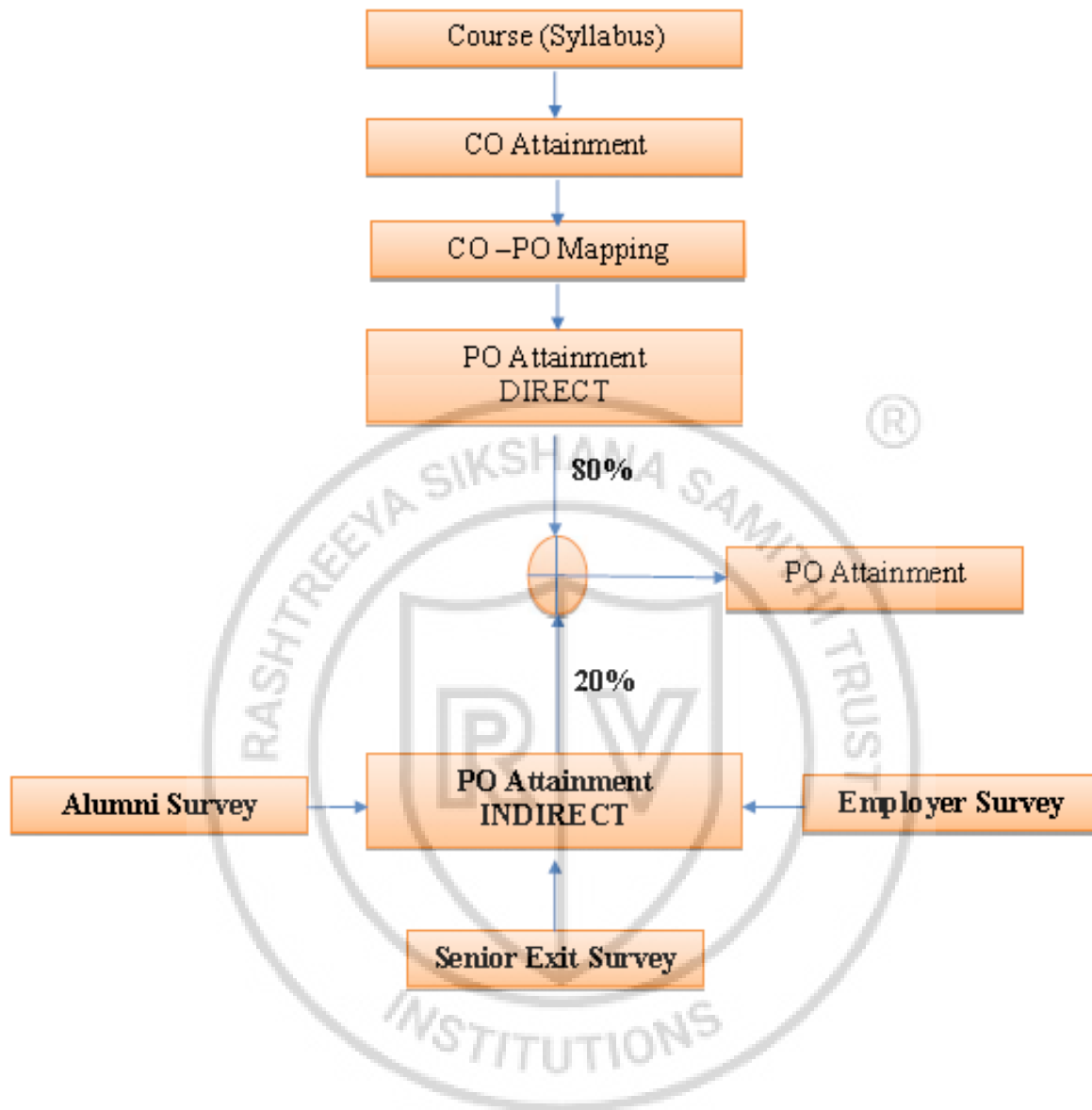
Process For Course Outcome Attainment



Final CO Attainment Process



Program Outcome Attainment Process





KNOWLEDGE & ATTITUDE PROFILE

- **WK1:** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- **WK2:** Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- **WK3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- **WK4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
- **WK5:** Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
- **WK6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
- **WK7:** Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
- **WK8:** Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
- **WK9:** Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.



PROGRAM OUTCOMES (POs)

- ❖ **PO1:** Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- ❖ **PO2:** Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- ❖ **PO3:** Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- ❖ **PO4:** Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- ❖ **PO5:** Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- ❖ **PO6:** The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- ❖ **PO7:** Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- ❖ **PO8:** Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- ❖ **PO9:** Communication: Communicate effectively and inclusively within the community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- ❖ **PO10:** Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- ❖ **PO11:** Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

INNOVATIVE TEAMS OF RVCE

Ashwa Mobility Foundation (AMF): Designs and fabricates Formula-themed race cars and mobility solutions to address urban transportation issues.

Astra Robotics Team: Focuses on designing and building application-specific robots.

Coding Club: Helps students gain coding skills and succeed in competitions like GSoC and ACM-ICPC.

Frequency Club Team: Works on software and hardware, emphasizing AI and Machine Learning.

Team Garuda: Develops a supermileage urban concept electric car and E-mobility products.

Team Jatayu: Builds low-cost UAVs with autonomous capabilities for various tasks.

Team Antariksh: Focuses on space technology and the development of operational rockets.

Team Chimera: Builds a Formula Electric Car through R&D in E-Mobility.

Helios Racing Team: Designs and tests All-Terrain Vehicles, participating in SAE's BAJA competitions.

Team Krushi: Creates low-cost farming equipment to assist farmers in cultivation and harvesting.

Team Vyoma: Designs and tests radio-controlled aircraft and UAVs.

Team Dhruva: Engages in astronomy-related activities and collaborates on projects with organizations like ICTS and IIA.

Ham Club: Promotes Amateur Radio and explores technical innovations in communications, especially for disaster response.

Chitrak : It is the official electric bike team of RV College of Engineering (RVCE) in Bangalore

Anoraniya: A Quantum based technical club initiated by students. The club is dedicated to addressing various technical challenges in quantum technologies and is on the verge of delivering some fascinating results.

Accelerate Club: organizes and participates in hackathons, bootcamps, workshops, and student-led innovation drives.

Cultural Activity Teams

1. AALAP (Music club)
2. DEBSOC (Debating society)
3. CARV (Dramatics club)
4. FOOTPRINTS (Dance club)
5. QUIZCORP (Quizzing society)
6. ROTARACT (Social welfare club)
7. RAAG (Youth club)
8. EVOKE (Fashion team)
9. f/6.3 (Photography club)
10. CARV ACCESS (Film-making)



NSS of RVCE



NCC of RVCE



VISION

Leadership in Quality Technical Education, Interdisciplinary Research & Innovation, with a Focus on Sustainable and Inclusive Technology



MISSION

- To deliver outcome based Quality education, emphasizing on experiential learning with the state of the art infrastructure.
- To create a conducive environment for interdisciplinary research and innovation.
- To develop professionals through holistic education focusing on individual growth, discipline, integrity, ethics and social sensitivity.
- To nurture industry-institution collaboration leading to competency enhancement and entrepreneurship.
- To focus on technologies that are sustainable and inclusive, benefiting all sections of the society.



QUALITY POLICY

Achieving Excellence in Technical Education, Research and Consulting through an Outcome Based Curriculum focusing on Continuous Improvement and Innovation by Benchmarking against the global Best Practices.



CORE VALUES

Professionalism, Commitment, Integrity, Team Work, Innovation



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